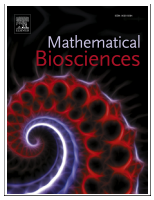
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Original research article

Reaction-diffusion models of invasive tree pest spread: Quantifying the expansion of oak processionary moth in the UK

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ABSTRACT

UK woodlands, forests, and urban treescapes are under threat from invasive species, exacerbated by climate change, trade, and transport. Invasive tree pests debilitate their host and disrupt forest ecosystems, thus it is imperative to quantitatively model and predict their spread. Addressing this, we model the spread of an invasive pest using a spatiotemporal reaction-diffusion equation, representing the spatial distribution as a population density field. We solve this intractable equation numerically and, from the solution, we determine first arrival times of the pest at locations in the field. The adopted model permits us to obtain the expansion rate of pest spread directly from the model parameters, which we infer in the Bayesian paradigm, using a Markov chain Monte Carlo scheme. We apply our framework to the ongoing spread of oak processionary moth in the UK, an outbreak which continues to grow despite management efforts. We demonstrate that our approach effectively captures the spread of the pest and that a model with a non-constant expansion rate is a better fit to the data than a constant-rate model. Finally, we discuss possible reasons for the apparent non-constant rate of expansion, arguing that this may be due to the impact of host species and environment heterogeneity, rather than a genuine change in the intrinsic expansion rate. The proposed framework is a powerful tool for quantitatively modelling the spread of an invasive tree pest and could underpin future prediction and management approaches.

1. Introduction

Invasive species pose a significant ecological, economical, and epidemiological threat both in the UK and globally [1,2], costing the UK economy over £ 4 billion per year [1]. The UK hosts over 100 invasive tree pests which were either introduced or have uncertain origin [3]. Climate change is anticipated to favour successful establishment of satellite populations and international trade is known to be a key driver of these introductions [3,4].

Imports of live oak led to the successful establishment of the oak processionary moth (OPM) in the South of England in 2006 [5]. OPM caterpillars – which defoliate and weaken trees – develop urticating setae which can shed and become airborne [6]. Human and/or animal contact with the caterpillars, their detached fibrous hairs, or their nests' detritus can lead to cases of caterpillar dermatitis [6–8]. Inhalation of the setae, which can break apart into microscopic fibres, can lead to respiratory irritation [8]. Since its establishment in the UK, OPM has continued to expand its range, from its original introduction in South-West London [9]. The climate in South England is already considered highly suitable for OPM, and Northern England and Scotland are expected to become suitable for OPM establishment by 2050 to 2070 [10].

Thus, there is urgency for a robust framework for modelling invasive tree pests and OPM in particular, to predict expansion and inform the government's ongoing surveillance and management programme [8]. OPM is a species of the order *Lepidoptera*, undergoing complete metamorphosis during its one-year development cycle [6]. The grey-white moths emerge between July and September, dispersing, mating, and ovipositing during an adult phase lasting 3 to 5 days [6]. The collective population reproduces and disperses in a single annual event [6]; any new nests in a given year are established by individuals who dispersed in the previous year. Estimates of the dispersal capabilities of adult OPM have been given as upwards of 20km, however this figure is for males [11]. Female OPM seldom nest far from their parent nest

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[6,12] and new nests are typically found within an average distance of 0.5km of the parent nest [12].

Recent efforts to quantify the spread of OPM in the UK have included the following. Townsend [13] and Cowley et al. [14] independently used amalgamated survey data from 2006 to 2012 to model the spread of OPM in the UK, the latter using electric network theory. In both cases, this (mostly) precedes the government's management strategy which was put in place in 2011, as well as the ongoing improvement to survey methodologies [9,15]. Similarly, a network approach was adopted by Wadkin et al. [15], with which the authors demonstrated the significance of kilometre-scale population dynamics on the population growth in Richmond and Bushy Parks – 'hot spots' for OPM presence in the UK. However, such network approaches require explicitly defined nodes, thus are challenging to scale to account for new sites or to model a continuous spatial domain. This work extended a previous characterisation of the population dynamics of OPM, which employed a stochastic susceptible-infested-removed model to estimate the time-varying infestation rate in the aforementioned parks [16,17]. Such compartmental models are often employed to study pest and pathogen spread: they account for both reproductive and dispersal dynamics [17,18], can be readily scaled to model arbitrary sized spatial domains, and capture time-varying phenomenon in the spread of a pest. However, as with the network approach, such models do not directly admit an expansion rate estimate from the model parameters.

The most recent expansion rate estimate for OPM spread in the UK was provided by Suprunenko et al. [9]. Applying a maximum distance method directly to the data, they characterised OPM spread by two phases of expansion 1) a slower phase, with population expansion at a rate of 1.66km/year from 2006 to 2014, followed by 2) a faster phase, at 6.17km/year, from 2015 to 2019 [9]. However, this approach does not classify the specific dynamics of the pest with a model that can be used to make forward predictions. As such, there is need to develop approaches to model and quantify the spread of OPM and invasive pests more generally, by incorporating their dynamics into a mathematical model capable of characterising time-varying spread and from which we can readily obtain an expansion rate for the pest.

In this work, we develop an approach to model the spread of OPM based on a reaction-diffusion equation with parameters inferred from observational data. Such equations encode both dispersal and reproductive dynamics, permit time-dependent parameters with which we can encode time-varying phenomenon, and admit an expansion rate directly from the model parameters. Reaction-diffusion equations have been applied to model populations of pine processionary moths [19], mosquitoes [20], plant-pathogen spread [21], and Neolithic humans [22], as well as the recent spread of COVID-19 [23]. Here, we represent the spatial distribution of the pest as a population density field which evolves according to a spatiotemporal reaction-diffusion equation. Adopting a fixed density threshold, we apply a binary mask to this density field to obtain a presence/absence field for OPM infestation. Consequently, we can determine (simulated) arrival times of OPM under a given parameter configuration. We adopt a Bayesian approach to inferring model parameters, employing an adaptive Markov chain Monte Carlo (MCMC) scheme to estimate the expansion rate of OPM in the UK [24].

The aims of this paper are threefold: (1) demonstrate the viability of a reaction-diffusion model of tree pest spread (and OPM in particular); (2) determine an estimate of the current expansion rate of OPM in the UK, and; (3) determine whether there is evidence that the spread of OPM has occurred in more than one phase.

2. Materials and methods

In this section, we describe the adopted approach of this paper. In Section 2.1, we present the OPM observational data used to train our reaction-diffusion models. In Section 2.2, we describe the reaction-diffusion model of OPM spread – the Fisher-Kolmogorov-Petrovsky-Piskunov (FKPP) equation [25,26]– along with our assumptions on the population. In Section 2.3, we define the two models of OPM spread proposed in this work. In Section 2.4, we outline a Bayesian approach to inferring the parameters of these proposed models. Finally, in Section 2.5, we provide the metrics used to validate these models.

2.1. OPM observational data

The training data (Fig. 1) are derived from amalgamated survey data (provided by the Forestry Commission and the University of Southampton), obtained from yearly surveys conducted between 2006 and 2023.

The reaction-diffusion model (see Section 2.2) captures the first arrival of OPM, but cannot capture the complex spatiotemporal processes behind the wavefront. We use a convex layer decomposition to estimate the average observed wavefront. Beginning with the final survey year $t = 2023$, we calculate the convex hull of (the locations of) the corresponding observed sites and interpret the vertex set of the hull as the observed wavefront in year t. We compare the sites observed in *previous* years to the observed wavefront and remove any that are located *ahead* of the wavefront. We attribute these observations to rare individuals who can fly further than the average individual, while the reaction-diffusion model is intended to capture the dynamics resulting from the latter. We set $t \leftarrow t - 1$ and repeat until (and including) the initial year $t = 2006$. Then, beginning at $t = 2006$, we construct the observed wavefront and compare this wavefront to sites observed in *later* years, removing those located *behind* the wavefront. These observations occur within the established range of OPM and thus do not contribute to the arrival of OPM in new territory under the reaction-diffusion model. We set $t \leftarrow t + 1$ and repeat until (and including) $t = 2023$. In Appendix A, we include Table A.3 which details the number of removed observations in each year, and a sensitivity analysis of the results obtained in Section 3 for the monophasic model to the removal of the 'ahead-of-front' observations. We found that the inclusion of these sites does not impact the inferred expansion rate. The remaining observations therefore consist of 414 locations and the years when OPM was observed at those locations. As we are interested in the *first* arrival of OPM in new areas, we retain only the earliest year OPM was observed at each location. These data are used as the training data for the models in Section 2.2. In Figs. A.8 and A.9, we present the yearly observations highlighting how many are included in the training data set in each year (see Fig. 2 for the retained observations in 2010, 2012, 2014, 2016, 2018, 2020, 2022, and 2023).

The observed data are point locations where OPM was first observed between 2006 and 2023, mapped to 1km^2 areas (see Fig. 1). These locations are provided as six-digit easting-northing Ordnance Survey National Grid coordinates which have units of metres. The areas are located within a $256\text{km} \times 256\text{km}$ square region Ω (panel (b) of Fig. 1). We discretize Ω to give a $\Delta\mathbf{x} = 1\text{km}$-resolution grid such that each $\mathbf{x}_{ij} \in \Omega$ is given by $\mathbf{x}_{ij} = (e_j, n_i)$, where e_j (resp. n_i) is the easting (resp. northing) coordinate of the centre of the corresponding 1km^2 area, with units of km. Thus, each of the 414 OPM observations consist of a location $\mathbf{x}_{ij}$ and a survey year t_k when OPM was first observed in the 1km^2 area with centre $\mathbf{x}_{ij}$. We refer to the pair $\mathbf{s}_k = (\mathbf{x}_{ij}, t_k)$ as the kth *observed site*, where $k = 1, \ldots, 414$. Note that the region Ω is large (compared to the extent of the observed data) to avoid boundary effects in our spatial simulations.

The initial (observed in 2006) sites corresponded to five observations scattered over a small region. In this work, we employ a reaction-diffusion model (see Section 2.2), which requires a continuous initial condition, and an analytic estimate of the expansion rate, the latter of which is valid provided the curvature of the wavefront is sufficiently small [28] (small curvature means little deviation from the tangent line, as when the distance of the front from the centre is sufficiently large). Following tests, we find the 2006 convex layer (which removes one of the five observations – see Appendix A) does not have sufficiently small curvature for the analytic expansion rate to be valid, identified by comparing cross-sections through x^c along the easting and easting-northing axes. Instead, we calculate the algebraic centre of mass for the

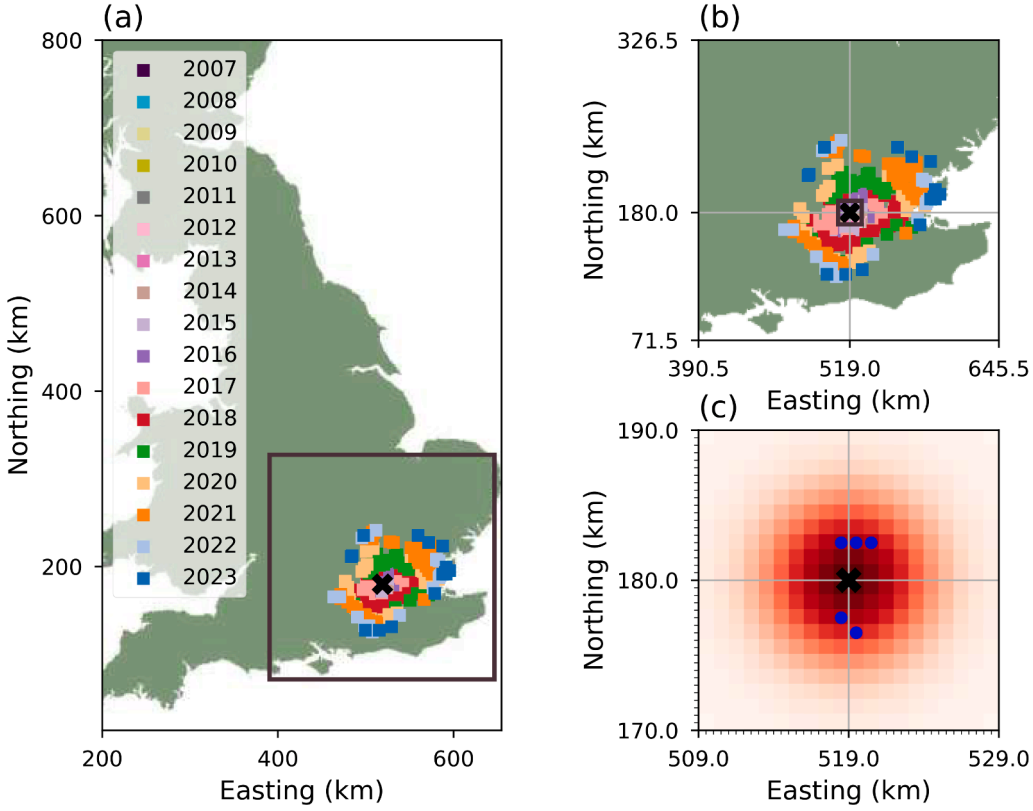


Fig. 1. The distribution of OPM nest observations. Each of the 414 coloured squares indicates a 1km^2 area where OPM is observed; the colour indicates the earliest year OPM was observed in that area. Black crosses indicate the centre of the initial distribution. Panel (a) shows the distribution on a map of the UK; (b) shows the 256km × 256km grid Ω adopted as the modelling domain (see Section 2.1), indicated by a square outline on panel (a); (c) shows the approximation of the initial (observed in 2006) distribution, with the true initial observations shown as blue dots. The minor ticks indicate the 1km resolution grid. For the background maps of the UK, we use the shapefile for Q. robur (English oak) provided by [27], normalised to one. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

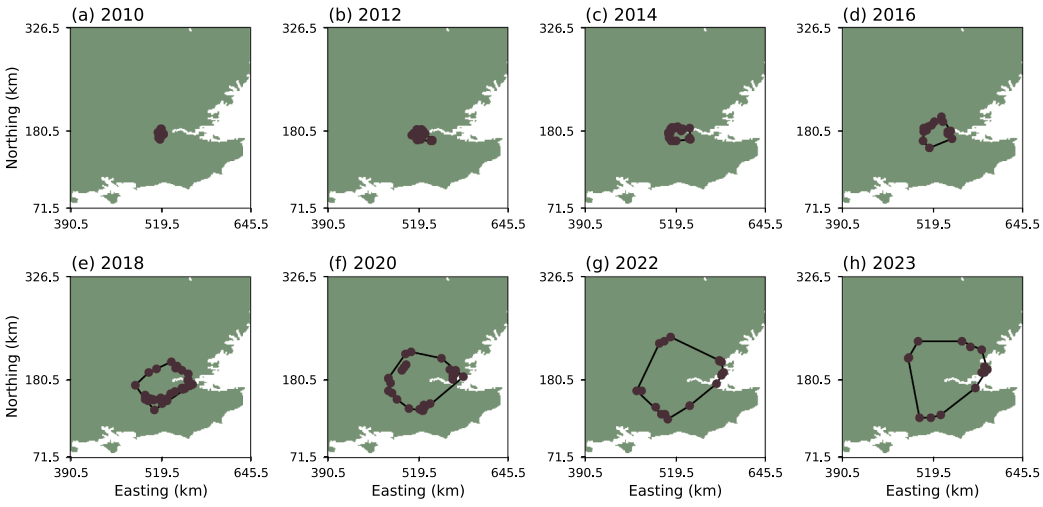


Fig. 2. The distributions of observed sites for (panel a) 2010, (b) 2012, (c) 2014, (d) 2016, (e) 2018, (f) 2020, (g) 2022, and (h) 2023. These are obtained by applying a convex layer decomposition (Section 2.1). For the background maps of the UK, we use the shapefile for Q. robur (English oak) provided by [27], normalised to one.

layered initial observations ($x^c = (519.0, 180.0)$) and the maximum distance $d = 3.54$km (2dp) of these sites from x^c, construct a circle with radius d and centre x^c, and finally apply a Gaussian blur to the circle, with standard deviation of 3.4. The value 3.4 was chosen as the smallest such value for which the wavefront evolves with sufficiently small curvature. This initial distribution is shown in panel (c) of Fig. 1.

2.2. FKPP equation for tree pest spread

In this work, we choose to quantify the spread of an invasive pest through a homogeneous landscape. This may be necessary for a new pest for which little is known about the impact of environmental factors or when such data is hard to come by. Additionally, available data may

come with uncertainty, thus an (over-)simplified model of pest spread could still be informative. For our application to OPM, we can validate the success of our approach against available estimates of the expansion rate [9].

We assume that the spread of an invasive pest is described by a population density field N given, at position $\mathbf{x}$ and time t, by $N(\mathbf{x}, t)$. We consider two modes of evolution (dispersal and reproduction) and that the host species is homogeneous in space (representing a lack of or inaccurate landscape distribution data). A simple assumption for dispersal through a homogeneous space is that such populations obey Fick's law for diffusion: populations move from regions of high density to regions of low density at a rate proportional to the population. Thus we assume dispersal of the pest is described by diffusion. A typically-assumed model of reproduction is logistic growth, where populations at low densities reproduce at an exponential rate which decreases as the density approaches some carrying capacity. Combining diffusion with logistic growth in the most simple way (adding them together), we obtain the FKPP equation (Eq. (1)).

The FKPP equation describes the evolution of a population density $N(\mathbf{x}, t)$ at position $\mathbf{x}$ and time t, given by

$$\frac{\partial N}{\partial t} = D\nabla^2 N + rN\left(1 - \frac{N}{K}\right), \tag{1}$$

with growth rate r, diffusivity D, and carrying capacity K [25]. The first term in Eq. (1) describes diffusion of N in space; diffusion represents dispersal of the population. The second term describes the logistic growth of N in time, where r is the growth rate and K the carrying capacity; logistic growth represents reproduction of the population subject to resource constraints encoded by K, the limit of the (local) population density. For constants r, D, and K (the so-called *isotropic* FKPP equation), one-dimensional solutions evolve to a propagating wavefront which travels at a speed of

$$v = 2\sqrt{Dr}, \tag{2}$$

if $N(x, t = 0)$ has compact support [26]. In two dimensions, this is true provided the curvature of the wavefront is small, such as when the distance of the wavefront from its source is large [28].

Here, we adopt units of years (yr) for t and units of kilometres (km) for (each component of) $\mathbf{x}$. While each individual moth disperses once per year, the population-level dispersal event is stochastic and believed to be tied to oak leaf burst, which occurs over several months [29]. Thus the precise time of individual dispersal will depend on local-level environmental factors; for simplicity, we take a continuous-time approximation for the population-level event. We assume the spatiotemporal distribution of the pest is described by the population field N and evolves according to the reaction-diffusion equation defined in Eq. (1); therefore, the density field evolves with a clear outward-travelling wavefront (see Fig. 3). For the parameters of Eq. (1), we assume a constant value of $K = 1$ for the carrying capacity and that the growth rate r and diffusivity D are uniform in space, but time-dependent. Additionally, we ignore potential impact from advective effects (e.g. prevailing winds) and human-mediated transport (e.g. vehicles on roads) for simplicity.

Eq. (1) has a natural lengthscale ℓ satisfying $D = \ell^2 r$, where r has units of 1/time (see e.g. [28]). The natural lengthscale is the OPM dispersal distance and ℓ has units of kilometres, so we set $\ell = 0.5$km, given in [12] as the expected distance between a new nest and its parent nest. The wavespeed $v = 2\sqrt{Dr}$ in Eq. (2) can be interpreted as the expansion rate of the population and has units of km/year. Since $\ell = 0.5$km, by Eq. (2), we have $v = 2\sqrt{\ell^2 r^2} = 2\ell r = (1\text{km}) \cdot r = r$ km/year (with $r > 0$, assumed implicitly in what follows). The specific forms we adopt for the growth rate r are given in Section 2.3.

We adopt a numerical approach to solving Eq. (1). For given growth rate r, we discretize Eq. (1) using 6th-order central finite differences with respect to the grid Ω (defined in Section 2.1), so that each ODE describes the time-evolution of OPM at one of the locations $\mathbf{x}_{ij} = (e_j, n_i) \in \Omega$. For simplicity, we assume Dirichlet boundary conditions at all four edges, fixed at zero. The boundary conditions are only a formality, since the spatial domain Ω is much larger than the extent of the training data and the wavefront never reaches the boundary in our simulations. We write $N^p = N(\cdot, t_p)$ for the state at time $t_p = t_0 + p\Delta t$. Starting at $p = 0$, the system is advanced from N^p to N^{p+1} by a step Δt according to a low-storage Runge-Kutta scheme [30]; we chose $\Delta t = 0.0125$ following convergence tests. We adopt the symmetric 3rd-order scheme defined in [31], which is parametrised by $a = [0, -2/3, -1]$, $b = [1/3, 1, 1/2]$, and $c = [0, 1/3, 2/3]$ (see [30] for notation).

At each time t_p, we apply a binary mask to the density field, i.e. $N^* = (N > N_T)$ such that $N^*_{ij} \in \{0, 1\}$, where N_T is the arrival threshold. Locations $\mathbf{x}_{ij}$ for which $N^*_{ij} = (N_{ij} > N_T) = 1$ are said to be infested with OPM at time t_p. We find all locations $\mathbf{x}_{ij}$ for which $N^*_{i,j} = 1$ *for the first time* (i.e. $N^*_{ij}(t_p) = 1$ and $N^*_{ij}(t) = 0$ for $t < t_p$) and identify these $\mathbf{x}_{ij}$ with the corresponding observed sites $\mathbf{s}_k$. We thus call t_p the *simulated arrival time* of OPM at these locations (see Section 2.4). For each site $\vec{s}_k$ for which t_p is the simulated arrival time under θ, we write $\tau(\vec{s}_k|\theta) = t_p$ to disambiguate the simulated arrival time from the corresponding observed time (Section 2.1). A schematic of this transition is shown in Fig. 3. Here, we adopt $N_T = 0.5$, since N_T is the midpoint between the two steady states $N = 0$ and $N = 1$, thus providing a reasonable estimate of the position of the front. It could be fruitful to investigate the impact of varying N_T, but we chose not to pursue this here in favour of reducing model complexity.

2.3. Proposed expansion rate(s)

We consider two models of OPM spread, each having the form of the reaction-diffusion equation described in Eq. (1). We denote by M_1 a model of the form Eq. (1) with

$$r(t) = r, \tag{3}$$

for some real value $r > 0$. The parametrisation of M_1 is $\theta_1 = (r)$, where we adopt vector notation for consistency (see definition of θ_2 below). This *monophasic* model M_1 encodes the assumption that the spread of OPM occurs at a constant rate r throughout the period 2006 to 2023. The second model M_2 is of the form Eq. (1) with

$$r(t) = \begin{cases} r_1 & \text{if } t \le t^{\text{critical}}, \\ r_2 & \text{otherwise}, \end{cases} \tag{4}$$

where $r_1, r_2 > 0$ are real values and the transition time t^{critical} satisfies $2006 \le t^{\text{critical}} \le 2023$. Hence, M_2 is parametrised by $\theta_2 = \left(r_1, r_2, t^{\text{critical}}\right)$. This *biphasic* model M_2 encodes the assumption that the spread of OPM occurs at a non-constant rate; at a rate of r_1 from 2006 up to and including t^{critical} and at a rate r_2 from t^{critical} to 2023. We provide snapshots of a prototypical numerical solution to this model in Fig. 3. We refer to these models as Model 1 (monophasic) and Model 2 (biphasic), respectively.

2.4. Parameter inference

Here, we outline a Bayesian approach to parameter inference for each of the proposed models (see Section 2.3).

We assume that the observed data $\mathcal{D}$ are generated by the proposed reaction-diffusion model(s) subject to spatially-independent Gaussian errors with spatially homogeneous standard deviation σ – the *observational noise parameter*. This captures expected local deviations from the idealised, global model - see e.g. [22]. These statistical assumptions permit a tractable Gaussian likelihood function given, up to proportionality, by

$$\pi(\mathcal{D}|\theta, \sigma) \propto \sigma^{-n} \exp\left[-\frac{1}{2\sigma^2}\sum_{k=1}^{n}\left\{t_k - \tau(\mathbf{s}_k|\theta)\right\}^2\right]. \tag{5}$$

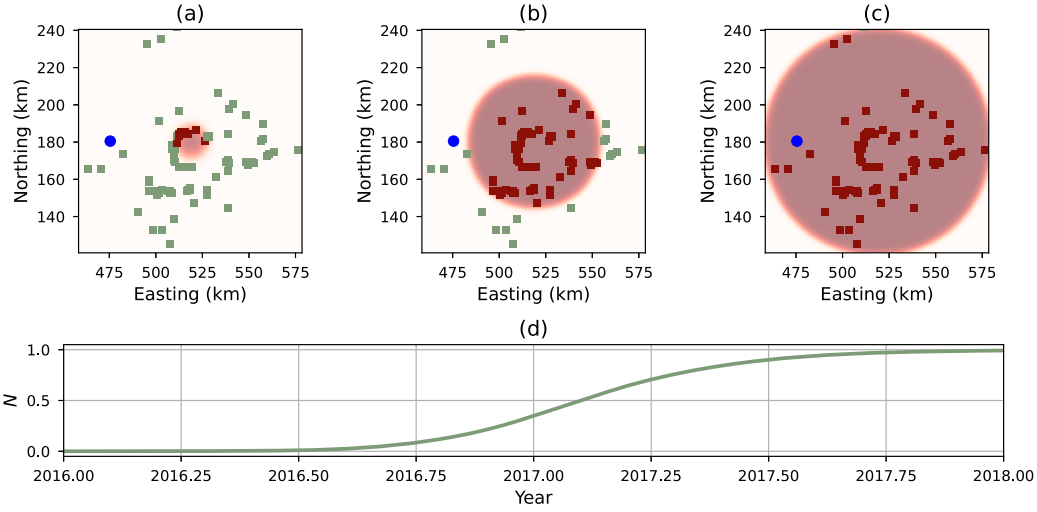


Fig. 3. A prototype wavefront solution. A prototype solution to Eq. (1) with $r(t)$ given by Eq. (4) with $r_1 = 0.35$, $r_2 = 6.21$, and $t^{\text{critical}} = 2014.17$ (shown in pink-red). This solution evolves from the initial distribution given in Section 2.1. Panel (a) shows $t = 2014$, panel (b) shows $t = 2018$, and panel (c) shows $t = 2022$. Red squares indicate sites that are infested at the corresponding time t (according to this prototype) and green squares indicate sites that are not infested. Panel (d) shows the time evolution of the density at the randomly-selected location (475.5, 180.5). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Inference proceeds in the Bayesian paradigm via the posterior distribution $\pi(\theta, \sigma|D)$ of the model parameters given the observed data, given by Bayes' theorem as

$$\pi(\theta, \sigma|D) \propto \pi(\theta, \sigma)\pi(D|\theta, \sigma). \tag{6}$$

Our initial beliefs about likely values for θ and σ are encoded by the prior density $\pi(\theta, \sigma)$. We adopt an independent prior specification and assume log-normal $\text{LogN}(1.73, 0.6241)$ distributions for each of the growth components, r in M_1 and r_1, r_2 in M_2. This specification corresponds to parameter values that are consistent with previous expansion rate estimates [9,13,32] and reduces bias towards a mono- or bi-phasic model. For the transition time component t^{critical}, we also adopt a truncated log-normal distribution with mean close in value to the average of 2006 and 2023; we take $t^{\text{critical}} \sim \text{LogN}(7.61, 0.006)$, with prior mean of 2014.65, standard deviation of 4.73 years, and a 95% credible interval of $(2006.2, 2022.6)$. Finally, we use an inverse Gamma distribution for the (square of the) observational noise parameter, so that $\sigma^2 \sim \text{IG}(2.23, 3.72)$. The prior variances are selected to be large enough to encode limited prior beliefs (thus making the likelihood more informative); we conduct a prior sensitivity analysis in Appendix B which suggests the results are insensitive to moderate increases in prior variance.

Due to the complex dependence of the numerical solution to (1) on the model parameters, the posterior is analytically intractable. Bayesian practitioners commonly use Markov chain Monte Carlo (MCMC) methods to sample from posterior distributions that are available up to an unknown constant of proportionality. Here, we adopt a particular MCMC scheme known as a Gibbs sampler [33]. In absence of analytically tractable full conditionals, a Metropolis-Hastings step can be used. Such an approach is termed a *Metropolis-within-Gibbs* scheme. Since the target posterior $\pi(\theta, \sigma|D)$ is analytically intractable, we sample from the full conditionals instead, alternating between draws of θ and draws of σ^2 (and σ) [22,33]. This procedure is defined as follows:

1. Initialise θ^0 and σ^0; set $j = 1$.
2. Draw $\sigma^j \sim \pi(\cdot|\theta^{j-1}, D)$.
3. Draw $\theta^{(j)} \sim \pi(\cdot|\sigma^j, D)$.
4. Set $j \leftarrow j + 1$ and return to step 2.

In step 3, the full conditional for θ is intractable, hence we adopt an adaptive Metropolis-within-Gibbs MCMC scheme to sample from the intractable posterior $\pi(\theta, \sigma|D)$ via a symmetric random-walk on a logarithmic scale to ensure all parameters are biologically feasible. Define $\lambda = \log(\theta)$, where the logarithm is applied component-wise. In step 3 of the Gibbs sampler, we propose a new value $\lambda^* = \lambda + \omega$, where $\omega \sim N(0, \Sigma)$, for a strictly positive-definite innovation matrix Σ – the covariance matrix of the proposal distribution. A proposal is accepted as the next value in the chain with a probability that ensures the invariant distribution of the Markov chain simulated by the Gibbs sampler is precisely the target distribution [24]. The choice of innovation matrix Σ influences the mixing of the Markov chain. It is common to adopt an *adaptive* MCMC scheme wherein the innovation matrix Σ is updated at fixed intervals to improve mixing [34]. We adopt the adaptive approach of Haario et al. [34]. The MCMC scheme is initialised with an arbitrary, strictly positive-definite, diagonal initial innovation matrix $\Sigma = \Sigma^0$. We select a *tuning period* $p = 10^4 \ll m$ retained iterations and a *tuning interval* $l = 100 \ll p$ where, for numerical convenience, we assume p is divisible by l. Let $\theta^{(j)}$ denote the jth value in the chain. For the first p samples, we update the innovation matrix at every l^{th} sample. Hence the innovation matrix for the jth step of the MCMC scheme is given by

$$\Sigma^j = \begin{cases} \Sigma^0 & \text{if } j < l, \\ \Sigma^{j-1} & \text{if } j/l \notin \mathbb{Z} \text{ and } j \leq p, \\ s_{d_k} \text{cov}(\theta^0, \ldots, \theta^{j-1}) & \text{if } j/l \in \mathbb{Z} \text{ and } j \leq p, \\ \Sigma^p & \text{otherwise,} \end{cases} \tag{7}$$

where the scaling parameter $s_{d_k} = (2.38^2)/d_k$ is a heuristic taken from [35] and d_k is the dimension of the parameter space of model M_k. For the initial innovation matrices Σ^0, we chose the prior covariance matrix with zero covariances for Model 1 – $\text{Diag}\{0.39, 5.6\}$ – and the diagonal matrix $\text{Diag}\{10^{-3}, 10^{-3}, 10^{-5}\}$ for Model 2 as these led to at least one acceptance in the first 100 steps and ensured the innovation matrix Σ^{100} was positive-definite. For a justification of this adaptive scheme, we refer the reader to [34]. The MCMC scheme above is iterated until m_T samples are obtained. We discard the initial $m_B \geq p$ samples as burn-in, and informally assess convergence of the remaining $m = m_T - m_B$ samples qualitatively using trace plots [24] and quantitatively using the effective sample size (ESS). We also apply additional scaling of the proposal variance to improve mixing [36]. See [22,24,33,34] for more details.

Table 1
Table of nine observed sites selected to present the posterior predictive sampling results. The fourth column d is the distance from $\mathbf{x}^c = (519.0, 180.0)$ to that point in kilometres (km); $\mathbf{x}^c$ is the centroid of the initial states (see Section 2.1).

Year	Northing (km)	Easting (km)	d (km)
2008	177.5	520.5	2.92
2010	174.5	518.5	5.52
2012	173.5	522.5	7.38
2014	178.5	509.5	9.62
2016	181.5	509.5	9.62
2018	161.5	532.5	22.90
2020	182.5	476.5	42.57
2022	187.5	583.5	64.93
2023	195.5	595.5	78.05

2.5. Model validation

We run the MCMC scheme for each of the models M_i described above. To compare the models, we use the *deviance information criterion* (DIC), which measures relative quality of models [37]. For model M_i (parametrised by θ_i and with observational noise σ_i), the DIC is given by

$$\mathrm{DIC}_i = 2\overline{D(\theta_i, \sigma_i)} - D(\bar{\theta}_i, \bar{\sigma}_i), \tag{8}$$

where $D(\theta_i, \sigma_i) = -2\log(\pi(D|\theta_i, \sigma_i))$ is the deviance of (θ_i, σ_i); this criterion quantifies goodness of fit while penalising over-fitting. The deviance is computed jointly over the (θ, σ) and with respect to all 414 observed sites. Models with a smaller DIC should be preferred to models with a larger DIC. We can readily compute $\bar{\theta}_i$, $\bar{\sigma}_i$, and $\overline{D(\theta_i, \sigma_i)}$ using the samples of (θ, σ), obtained from the (final) MCMC run. Finally, we assess goodness of fit of the model with lowest DIC using *posterior predictive sampling*. Let $\tilde{t}_k$ denote a first arrival time at site $\mathbf{s}_k$. The predictive density is given by

$$\pi(\tilde{t}_k|D) = \iint f\left(\tilde{t}_k|\tau(\mathbf{s}_k|\theta), \sigma\right)\pi(\theta, \sigma|D)\,\mathrm{d}\theta\mathrm{d}\sigma, \tag{9}$$

where f denotes the Gaussian density with mean $\tau(\mathbf{s}_k|\theta)$ and variance σ^2. Hence, given samples $\left\{\left(\theta^{(j)}, \sigma^{(j)}\right)\right\}_{j=1}^{m_p}$ from $\pi(\theta, \sigma|D)$, we generate samples $\tilde{t}_k^{(j)}$ from $\pi(\tilde{t}_k|D)$ by drawing from $\tilde{t}_k^{(j)} \sim N\left(\tau\left(\mathbf{s}_k|\theta^{(j)}\right), \left(\sigma^{(j)}\right)^2\right)$ for $j = 1, \ldots, m_p$. Thus, under the preferred model, we generate $m_p = 10^4$ predicted observations for each of the 414 observed site. We select nine survey years spaced across the whole survey period. From each of these years, we calculate the distance (from the centre $\mathbf{x}^c$) of all sites observed in that year and select the site with distance closest to the median distance. These nine selected test sites (given in Table 1) are selected for visualisation only as they provide reasonable spatial coverage of the observed sites; we show the posterior predictives at these nine sites in the boxplots in Figs. 6 and 7.

3. Results

For each model, we run the adaptive MCMC scheme to obtain at least 5.0×10^4 initial samples. During these runs, adaptive tuning was applied every 100 samples until the $(10^4)^{\text{th}}$ sample was obtained. These runs were followed by batches of 10^4 iterations of MCMC until satisfactory convergence and mixing. We then retained only the final 3×10^4 samples from each run. We provide the resulting trace plots in Appendix C, which provide a visual diagnostic check of adequate mixing.

Recall that Model 1 is parametrised by $\theta_1 = (r)$ and σ_1. We initialise the MCMC scheme with values $\theta_{1,0} = 4.0$ and $\sigma_{1,0} = 2.0$, chosen arbitrarily. After burn-in, we obtain 3×10^4 samples from the posterior distribution of the parameters of Model 1; this is the length of the final Markov chain. This output is summarised by Fig. 4; the trace plots (Fig. C.10) suggest satisfactory convergence of the MCMC scheme, as do the effective sample sizes (ESS) given in Table C.6. In Table 2, we provide a summary of the inferred parameters for each model. For Model 1, with a constant expansion rate, we obtain a posterior mode of $\hat{r} = 2.68$ with 95% (equi-tailed) credible interval (CI) $(2.56, 2.81)$ for the growth parameter r. For the observed noise parameter σ, the posterior mode is $\hat{\sigma}_1 = 4.81$ with 95% CI $(4.50, 5.15)$. The DIC reported for this model is $\mathrm{DIC}_1 = 1720.74$. All summary statistics are given to two decimal places.

Table 2
Table of results for the reaction diffusion models. We provide, for each of the two proposed models, the calculated deviance information criterion (DIC) as well as the mean, 95% equi-tailed credible interval (CI), sample size (SS), and effective sample size (ESS) for each of the model parameters.

Model	$\hat{\cdot}$	Mean	95% CI	SS	ESS	DIC
1	r	2.68	(2.56, 2.81)	3×10^4	1803.75	1720.74
	σ_1	4.81	(4.50, 5.15)	3×10^4	28508.55	
2	r_1	0.32	(0.29, 0.35)	3×10^4	2622.82	831.60
	r_2	6.14	(5.77, 6.56)	3×10^4	2390.97	
	t^{critical}	2014.07	(2013.74, 2014.39)	3×10^4	2696.18	
	σ_2	1.65	(1.54, 1.77)	3×10^4	28486.24	

Model 2 is parametrised by $\theta_2 = \left(r_1, r_2, t^{\text{critical}}\right)$ and σ_2. We initialise the MCMC scheme with $\theta_{2,0} = (4.0, 4.0, 2010.0)$ and $\sigma_{2,0} = 2.0$; initial values for the parameters in common with Model 1 are chosen as above and the remaining values are chosen arbitrarily. We obtain 3×10^4 samples from the posterior of Model 2; This is the length of the final Markov chain (see Figs. C.11, 5, and Table 2). For this model of a non-constant expansion rate, we obtain posterior modes of $\hat{r}_1 = 0.32$ (CI: $(0.29, 0.35)$), $\hat{r}_2 = 6.14$ (CI: $(5.77, 6.56)$), $\hat{t}^{\text{critical}} = 2014.07$ (CI: $(2013.74, 2014.39)$), and $\hat{\sigma}_2 = 1.65$ (CI: $(1.54, 1.77)$). The DIC for this model is $\mathrm{DIC}_2 = 831.60$ (2dp).

The values for the DICs of Models 1 and 2 (Table 2) indicate that Model 2 is the more appropriate choice for the observed data of the two models considered. This suggests that a biphasic expansion model is more appropriate for describing the spread of OPM than a monophasic model. For each of the models, we generated m_p posterior predictive samples and present box plots in Figs. 6 and 7 summarising the posterior predictive distributions at selected observed sites (see Section 2.1). It is clear that both models are able to produce predictions that are consistent with the observations, with most data points lying within the interquartile range (IQR) of the corresponding predictive distribution, and all within a factor of 1.5 of the IQR. In particular, the predictions of Model 2 in the later years (2022 and 2023) are better than those produced by Model 1.

4. Discussion

In this section, we begin with comments pertaining to the interpretation of these results and model validity; in Section 4.1, we discuss the implications of these results for the spread of oak processionary moth in the UK, finishing with ecological considerations. In Section 4.2, we comment on the limitations of the adopted modelling approach. Finally, in Section 4.3, we highlight some potential directions for future work.

4.1. The spread of oak processionary moth in the UK

The values for the DICs of Models 1 and 2 (Table 2) indicate that Model 2 is the more appropriate choice for these observed data. This could suggest that the expansion of oak processionary moth in the UK from 2006 to 2023 occurred at a non-constant rate. This conclusion aligns with previous results in the literature [9]. However, we note later in this discussion that this apparent biphasic expansion could be attributable to landscape and environment heterogeneity or changes in management strategy, and not a change in the intrinsic rate of expansion.

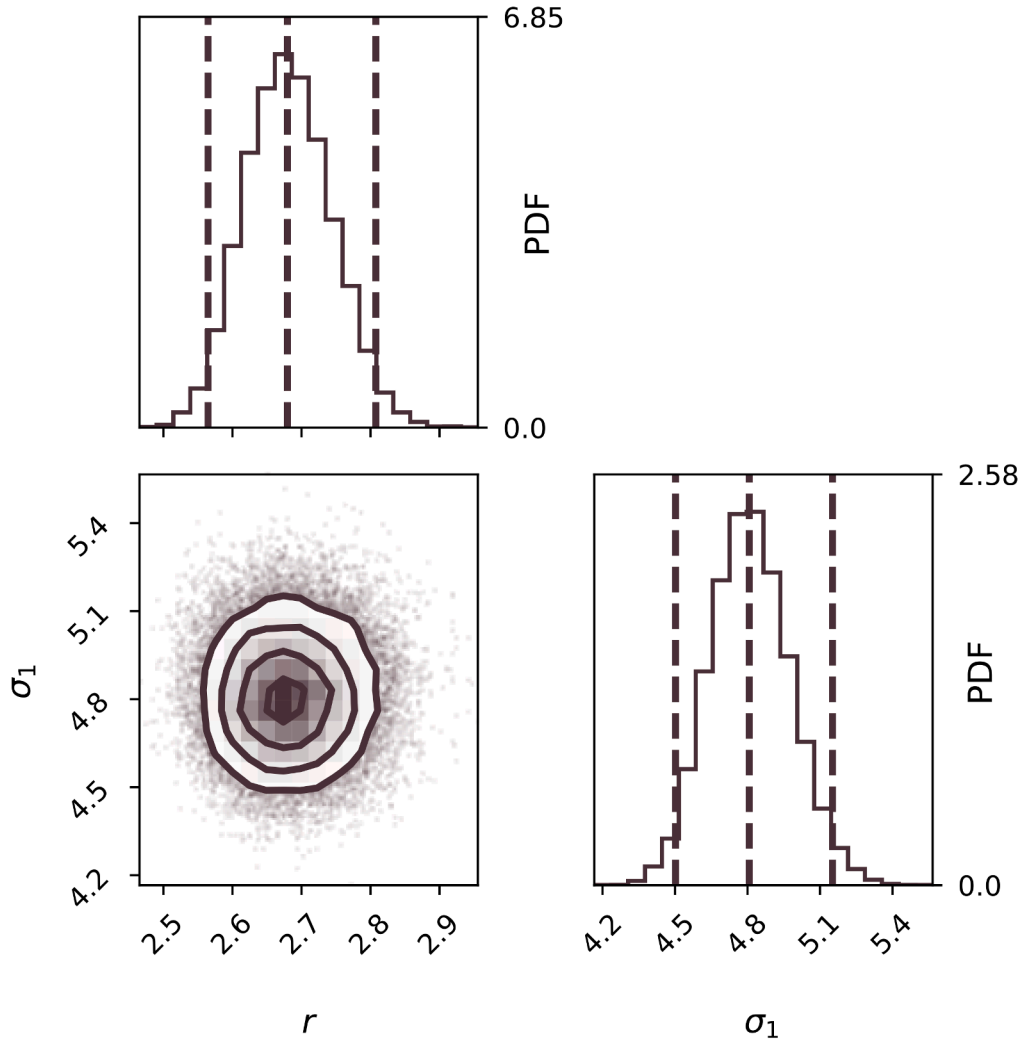


Fig. 4. Model 1 posterior distributions. Posterior distributions (histograms) and kernel density estimate of the joint posterior (contour plot) for r and σ_1, using 3×10^4 samples from the MCMC scheme. (Dashed) lines indicate the posterior mode and 95% credible interval.

The posterior for Model 1 suggests an expansion rate that is lower than estimates reported in the literature [9,11,13,32]. However, we use survey data for the full period 2006 to 2023; no previous work determines expansion rates using data including the period 2019 to 2023. These results could suggest that the rate of expansion from 2019 to 2023 was lower than the expansion rate from 2006 to 2019. Additionally, we note that the posterior indicates a modal observed noise parameter of 4.82 years; this is fairly large with respect to the 18-year period these data cover and suggests the constant-rate model does not accurately estimate the arrival of OPM at the observed sites (on average). This could be due to the prescribed initial state for the distribution of OPM in 2006 (see Section 2); a number of observed sites are located where the corresponding initial state N^0 is non-zero (see Section 2.2), thus our model may obtain earlier arrival times for these sites than the true values, irrespective of model parameter choices. We explored this further in Appendix A. Inferences for both models were insensitive to employing a tighter initial Gaussian profile and those for the monophasic Model 1 were insensitive to the removal of the overlapping points. However, we found the biphasic Model 2 inferences were moderately sensitive to their removal; nonetheless, the 95% credible intervals were reasonably close despite no overlap and, we still inferred the stark r_1 vs r_2 contrast obtained when these observations were retained. To ensure consistent data use across both models, we elected to retain these observations.

The posterior for Model 2 suggests that the apparent biphasic spread occurs at (posterior mode) rates of $\hat{r}_1 = 0.32$ and $\hat{r}_2 = 6.14$, with (posterior mode) phase transition time $\hat{t}_c = 2014.07$. Since the MCMC for Model 2 is initialised with the same initial growth rates and corresponding priors, matching those of Model 1, the difference between the phase 1 and phase 2 growth rate posteriors gives a strong indication that the data prefers a model with non-constant expansion rate over one with a constant expansion rate. This conclusion is further supported by the DICs. Box plots of the posterior predictive observations are provided in Fig. 7. With the exception of 2012, 2016, 2018, and 2023, all true observations lie within the interquartile range of the posterior predictions for that year. We explored whether these erroneous predictions may result from the removal of observations via convex layering (Section 2.1) during model development, finding that the inferred expansion rate for the monophasic Model 1 is relatively insensitive to the removal of these observations. Again, to ensure consistency, we elected to use the layered data as training data for both models (see Appendix A).

The results for the model with a non-constant rate of expansion align with previous results, which also indicated two phases of expansion (at rates 1.66 and 6.17, respectively) with transition time 2014 [9]. We note that the weighted average of $\hat{r}_1$ and $\hat{r}_2$ (3.23) is reasonably close in value to the Model 1 growth rate (2.91), and closer still to the mean of the previous biphasic estimates (3.27) [9].

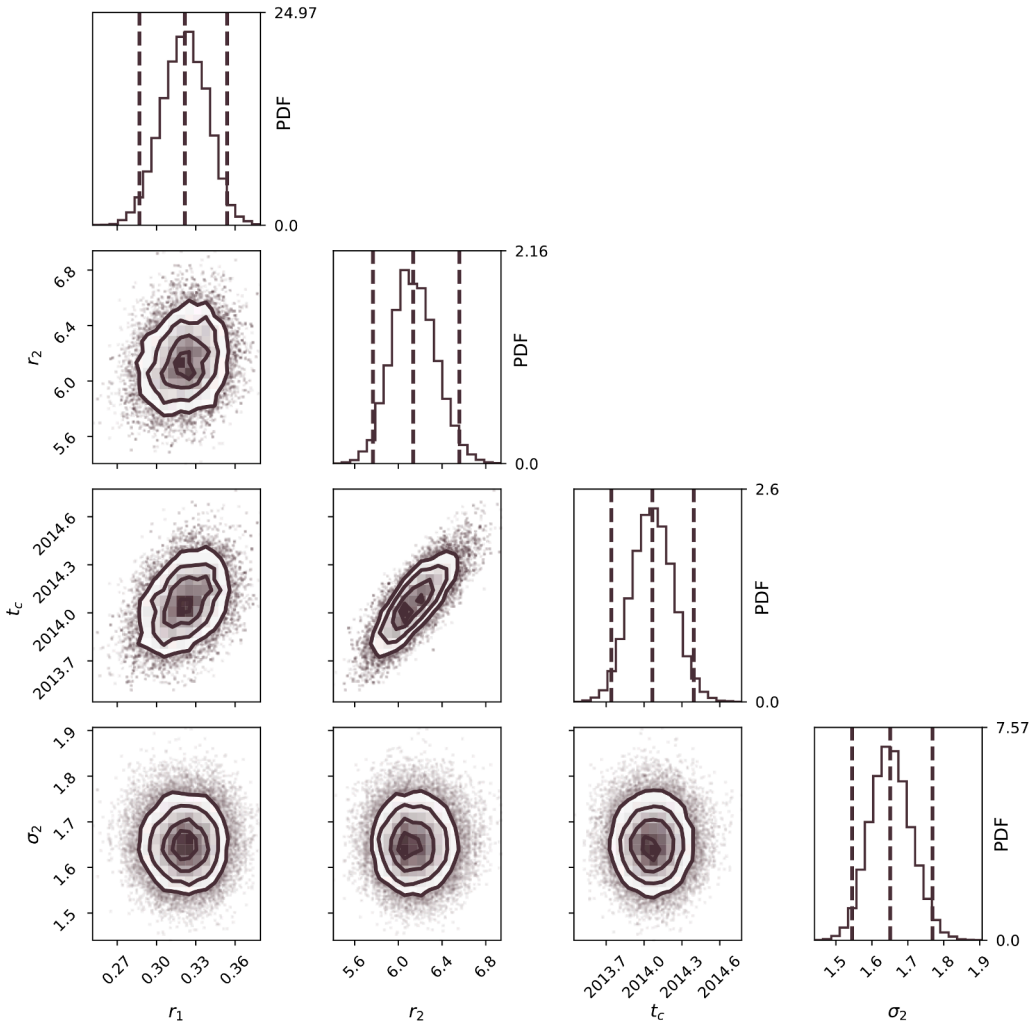


Fig. 5. Model 2 posterior distributions. Posterior distributions (histograms) and kernel density estimates of the pairwise marginal posteriors (contour plots) for r_1, r_2, t^{critical}, and σ_2, using 3×10^4 samples from the MCMC scheme. (Dashed) lines indicate the posterior mode and 95% credible interval.

Realistically, the carrying capacity, growth rate, and diffusivity will all depend on space and time due to a range of factors not accounted for by this model, such as weather, seasonality, and host density and distribution [9,38,39]. Thus, ecologically, a constant rate of expansion could be deemed unrealistic; the spread of a pest is likely influenced by landscape and climatic variability [6,12,38]. These factors could be time-dependent; seasonality results in cyclic temperature trends and consumption of woodland by society (e.g. via the forestry industry) varies with demand and thus possibly with time. Additionally, it was noted in [40] that some invasive populations can initially decline before slowly increasing over several years. This could manifest in terms of an initial slower phase of expansion, in which a declining population results in fewer new 1km^2 areas where the pest is observed. A second phase of expansion, characterised by a higher expansion rate, could follow due to the adaptive evolution of the pest to its new environment [12,41], which is expected to occur over several generations [41]. Alternatively, the apparent biphasic spread of OPM could be the result of spatial variability, in landscape and climate, that are known to impact the spread of a pest [6,12,38]. Furthermore, we model dispersal as diffusion; spatially-explicit diffusion can be incorrectly interpreted as time-dependence [42]. Finally, the management programme could influence the apparent biphasic spread. Initially, management was focussed on aggressive early control and eradication, changing later to a longer-term containment strategy around 2014 thus some of the observed apparent acceleration in the second phase could result from increased sampling intensity or more proactive searching ahead of the front in later years.

4.2. Modelling approach

We made a number of simplifications in our approach to capture the spread of oak processionary moth. Firstly, we assumed a Gaussian initial condition for the models (see Section 2.1). In this work, we treated the OPM distribution in each year as a single macroscopic population, including the initial distribution since the initial sites are close together (within a circle of radius 3.54km). We took the mean of the Gaussian as the algebraic centre of the initial observations. This assumes the observations contribute equally to the centre of mass calculation, as nest number data is inconsistently and inaccurately reported. Likewise, the fixed width of the initial distribution was chosen as the smallest such width where all initial observations were contained within its extent and the initial profile had sufficiently small curvature for the analytic expansion rate to be applicable. Although it would be possible, in principle, to infer the initial condition as part of the inference scheme, we chose not to pursue this. A larger width would assume a larger initial range than is recorded, while a smaller width would not contain all initial observations. We explore the impact of decreasing the width of the initial profile and on removing observations which overlap with it in

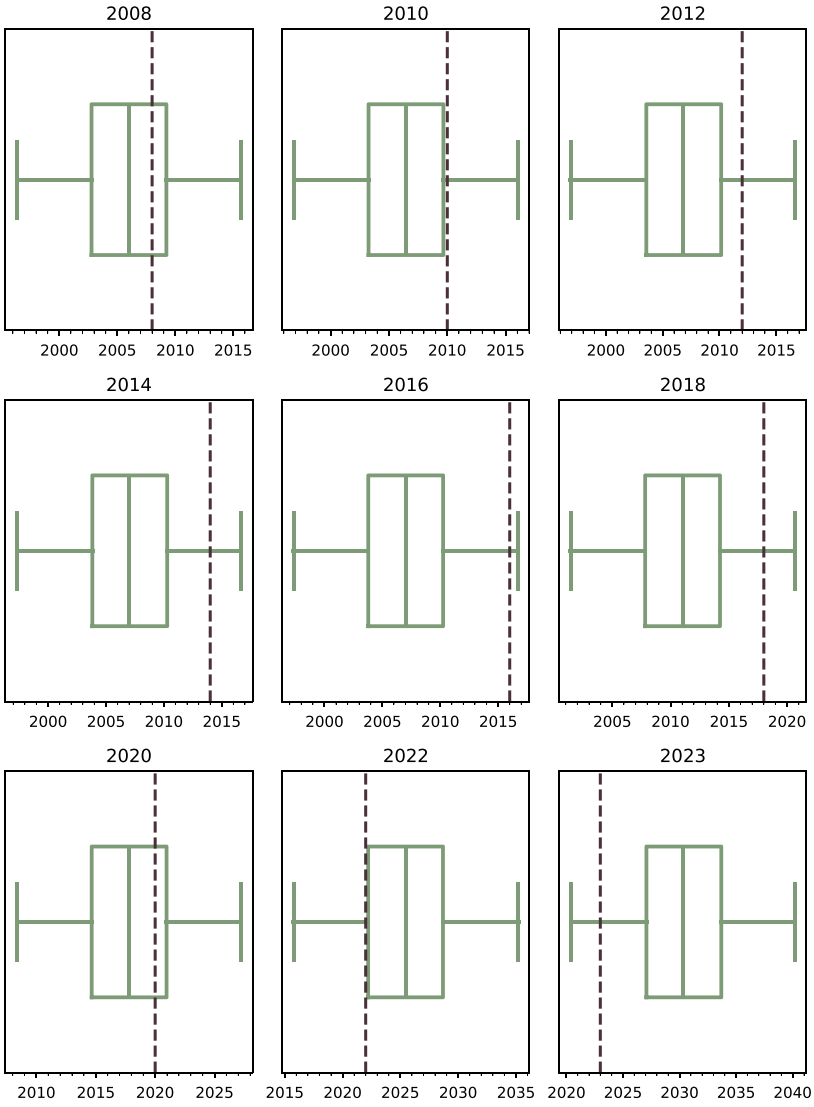


Fig. 6. Model 1 posterior predictives. Box plots summarising the predictive distribution at nine observed sites (test sites) (see Table 1). Whiskers indicate ±1.5IQR. Observations are indicated by a vertical dashed line, with the corresponding observed year above the axis.

Appendix A. Secondly, the data has a distinct asymmetry in later years, with evident preferential spread towards the North-East (see e.g. [9]). Thus it is likely the inferred mean will be dominated by these observations and possibly result in an initial distribution that does not contain the initial observations present in the survey data.

The second major simplification is in the adopted form of the model, that of a homogeneous/host-density independent spread under simple diffusion. One may argue that a simpler piecewise-linear radial model could capture the same pattern of spread as captured by the reaction-diffusion model. We explore this in Appendix D. While landscape structure is expected to impact the spread of a pest, we have demonstrated that our simple approach adequately describes the spread of OPM in the UK. Additionally, the reaction-diffusion model allows incorporation of pest dynamics (dispersal and reproduction) as individual terms, allowing us to account for the impact of both on the overall spread and, in the case of the fixed lengthscale $\ell = 0.5$, account for the apparent dominance of local reproduction on OPM spread (see below discussion of the lengthscale). Such a classification would not be readily obtainable from a radial model, thus the reaction-diffusion model is more amenable to relaxations of the assumptions we adopt in this work.

Previous estimates of the expansion rate of OPM in the UK were most recently obtained using a maximum distance method [9]. With our reaction-diffusion approach, we can obtain an estimate of the rate directly from our model parameters. This offers benefits; this framework can readily estimate future expansion rates. However, the equation for the expansion rate (Eq. (2)) is obtained analytically, hence it will be susceptible to error introduced by the choice of numerical integration method. Fortunately, testing indicates acceptable convergence of the numerical scheme for our model, spatial-, and temporal-scales. Additionally, this analytic expansion rate is an asymptotic limit and may not be accurate in the earlier years of spread, and it requires a circularly-symmetric initial profile to be applicable for in all years.

Our spatially-explicit modelling approach can provide an estimate of the arrival of OPM at every 1km area in a 256km × 256km spatial domain and can be readily extended to larger domain sizes, provided adequate computational resources are available. Our reaction-diffusion equation describes the evolution of the population density field; we apply a binary mask to this to facilitate comparison of our model with observed presence/absence data. Additionally, this could facilitate direct comparison of our approach with similar (compartmental) approaches [17,18]. Furthermore, the population density field evolves with a clear

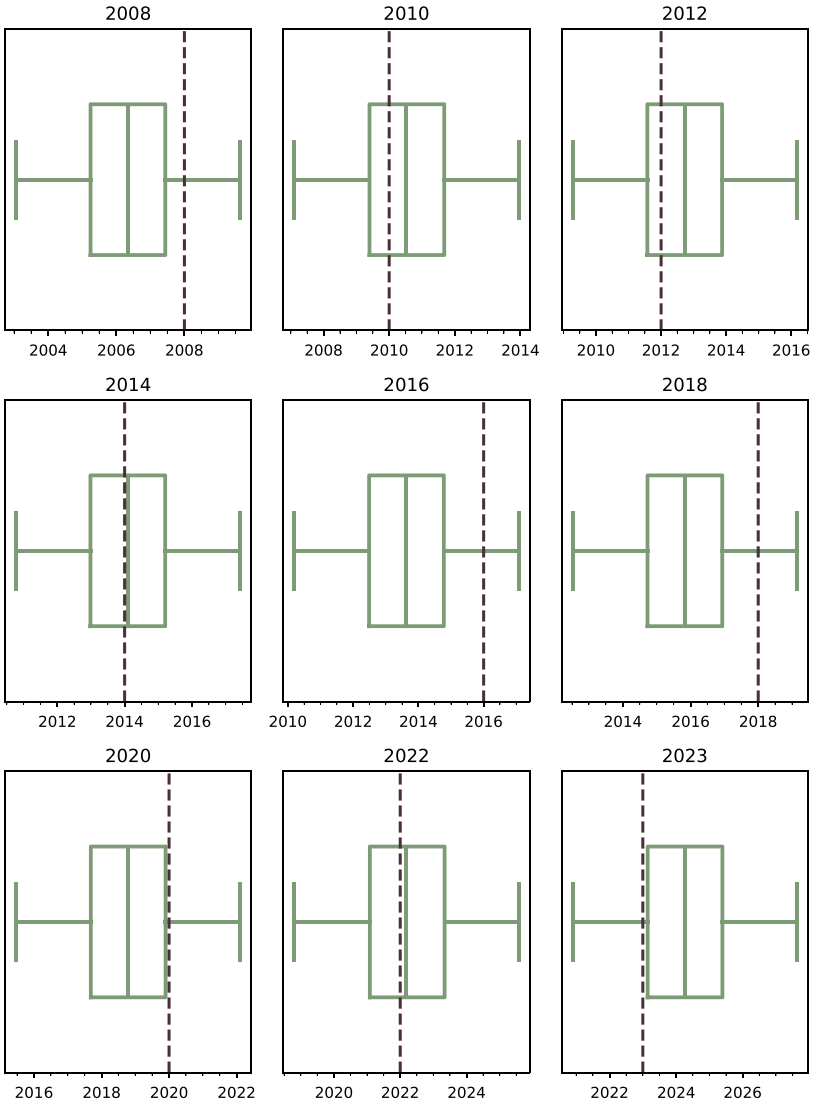


Fig. 7. Model 2 posterior predictives. Box plots summarising the predictive distribution at nine observed sites (test sites) (see Table 1). Whiskers indicate ±1.5IQR. Observations are indicated by a vertical dashed line, with the corresponding observed year above the axis.

outward-travelling wavefront (as in Fig. 3). This captures the spatial trend of first arrival of the pest without incurring the computational cost of simulating a large number of individuals, for example. The use of a binary mask also allows our model to somewhat account for potential time lag in observation. Time lag is posited to reflect the time needed for populations to reach detectable densities [43].

The adopted form of a reaction-diffusion equation permits calibration of the dynamics equation (Eq. (1)) to modelling OPM by fixing a specific lengthscale ℓ, taken from available estimates of the expected dispersal distance of a (female) OPM in a single dispersal event [12]. This facilitates application of our framework to other invasive pests, provided suitable spatial and temporal scales can be adopted and a reasonable estimate for ℓ obtained. This lengthscale ℓ determines the motility of the population with respect to its (local) reproduction, thus results may be highly sensitive to inaccuracies in estimating ℓ. This could be challenging for problems where the primary goal is to determine risk of infestation in previously uninfested areas. Our adopted lengthscale for OPM is $\ell = 0.5$ [12]; most adult females probably nest between 0 and 1km from where they emerge and most likely 0.5km [12], hence our choice here is reasonable. We note that other estimates are available, ranging from 5 to 20km [11,12,32], but these estimates have been attributed to extreme female flight capabilities.

Reaction-diffusion models, such as the one considered here, require specification of several parameters, resulting in analytically intractable posterior distributions. In this work, we adopt a Bayesian approach to parameter inference, via computation of the joint posterior distribution over all parameters of interest, using data consisting of first arrival times at some 414 locations across the South of England. We assume that the true underlying arrival time is not observed exactly, but subject to Gaussian noise, giving a tractable likelihood function. However, the posterior distribution is intractable, necessitating the use of sampling-based approaches such as MCMC [24]. In particular, these are routinely employed to infer parameters for models of epidemics [15–17] and collective behaviour [22,44]; the adopted scheme requires the likelihood to be known only up to proportionality. Specifically, we adopt a Metropolis-within-Gibbs scheme; such approaches are popular among ecological population modellers [15–17,22,34,44]. We inferred parameters of the models using an adaptive random-walk Metropolis-within-Gibbs approach to generate posterior samples. We also note that, while the observations are discrete (integer survey year) arrival times bounded between 2006 and 2023, the ascribed Gaussian likelihood places mass on

non-integer arrivals as well as those outside the survey range. It may be more appropriate to instead employ a likelihood with discrete support, but we chose not to pursue this. For model comparison, we use the deviance information criterion to assess the relative quality of the two competing models. This metric penalises models based on goodness of fit and the effective number of parameters, and is routinely applied when fitting models in the Bayesian paradigm [37]. Other metrics can be computed, such as the (frequentist) Akaike information criterion (AIC), which requires maximising the likelihood to obtain the maximum likelihood estimate, rather than estimating this directly from the sample average. As an exercise, we calculated the AIC and found this also indicated Model 2 as the preferred model.

4.3. Further work

Our work, in particular, could indicate that the spread of OPM in the UK occurred in two phases, but further work would illuminate whether there are more phases of expansion. This apparent time-dependence could be misleading; it is known that spatial dependence of the diffusivity D can be misinterpreted as time-dependence [42]. Furthermore, dispersal capabilities and population resource constraints are highly likely to depend on environmental factors, such as temperature, precipitation, and host distribution [6,12,38]. Consequently, we believe that environmental factors – in particular dependence of spread on host distribution – are the likely reason for the apparent biphasic expansion of OPM in the UK. In this work, we adopt a reductionist approach [45], whereby we omit dependence on landscape and environmental factors in favour of a simpler model. Further study is needed to understand how host density affects OPM spread and how this is best incorporated into a reaction-diffusion model. We note that recent work used a coupled compartmental-dispersal model to predict the effect of landscape structure on epidemic invasions of agricultural crops [18]. Also of interest is the impact of undetected presences and a spatially-varying carrying capacity (e.g. landscape variability), though such investigations will necessitate non-trivial modifications to the modelling and inference framework.

CRediT authorship contribution statement

Jamie P. McKeown: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization; **Laura E. Wadkin:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization; **Nick G. Parker:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization; **Andrew Golightly:** Writing – review & editing, Methodology; **Andrew W. Baggaley:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors of this work have no competing interests to declare.

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Data availability

The observational OPM nest data is available from the Forestry Commission, England. Background maps of the UK used in Figures 1 and 2 are derived from oak abundance data in [27], available at [46]. Code and data produced by this work are available at [47].

Appendix A. Sensitivity to the data parsing

In the data cleaning and parsing regime presented in Section 2.1, we first apply a convex layer decomposition to isolate the OPM observations which can be attributed to natural spread and remove those attributable to secondary infestations ('behind-the-front') and infrequent long-range dispersals or human-mediated spread ('ahead-of-the-front'). Since the adopted reaction-diffusion models capture the invasion front propagating into new territory, behind-the-front observations do not provide additional insight into the expansion of OPM's territory and can be removed from the training set. Figs. A.8 and A.9 summarise the spatial distribution of removed and retained observations in each year (see also Table A.3 for a year-by-year breakdown of the removed points).

Table A.3
Table of yearly OPM observations. We provide, for each year of observation, the number of 1km^2 observations for which we use only the first arrival times as training data for our models (Column 1), the number of observations removed due to appearing within the observed territory of OPM in some earlier year (Column 2) or outside of the observed territory in some later year (Column 3), and the total number of removed observations (Column 4).

Year	Retained	Removed: behind	Removed: ahead	Total Removed
2006	4*	0 (0%)	1 (100%)	1
2007	4	6 (75%)	2 (25%)	8
2008	3	12 (75%)	4 (25%)	16
2009	5	14 (93%)	1 (7%)	15
2010	22	28 (93%)	2 (7%)	30
2011	0	2 (33%)	4 (67%)	6
2012	45	36 (86%)	6 (14%)	42
2013	16	119 (> 99%)	1 (< 1%)	120
2014	35	162 (99%)	2 (1%)	164
2015	48	172 (100%)	0 (0%)	172
2016	15	182 (98%)	4 (2%)	186
2017	33	338 (100%)	0 (0%)	338
2018	41	535 (> 99%)	2 (< 1%)	537
2019	34	526 (> 99%)	1 (< 1%)	527
2020	29	885 (> 99%)	6 (< 1%)	891
2021	51	973 (> 99%)	4 (< 1%)	977
2022	17	980 (> 99%)	6 (< 1%)	986
2023	16	605 (100%)	0 (0%)	605
Total	418	5574	46	5620

* The four retained sites in 2006, used to construct the initial condition, are not part of the set of 414 observations used as training data for the models.

During model development, we verified that the inferred expansion rate for the monophasic model is robust to the removal of the ahead-of-front observations. Secondly, we took as the initial conditions of our models a Gaussian profile centred on and encapsulating the initial observations. In the Discussion, we argued that the width of the Gaussian and the presence of sites overlapping with non-negligible initial density could result in earlier arrivals at these sites, irrespective of model parameter choices. This would manifest in a larger observational noise parameter. Below we summarise the results of investigating the impact of these two choices and our hypotheses of their effects.

We consider three additional configurations of the data. In configuration (a), we adopt a tighter Gaussian initial condition for both models, halving the radius of the circle fitted to the initial states prior to applying a Gaussian blur. In configuration (b), we define non-negligible initial density to be a quarter of the maximum initial density ($N \approx 0.11$) and remove any observations at locations with this density or higher. Configurations (a) and (b) are considered for both the monophasic and biphasic

Table A.4
Table of results for sensitivity analyses of model outputs to the data parsing regime. For each model, configuration, and component parameter chain, we provide the mean, 95% equi-tailed credible intervals (CI), sample size (SS), and effective sample size (ESS).

Model	Config.	$\hat{\cdot}$	Mean	95% CI	SS	ESS
1	(a)	r	2.78	(2.67, 2.91)	3×10^4	4201.16
		σ_1	4.44	(4.15, 4.76)	3×10^4	29097.39
	(b)	r	2.78	(2.68, 2.89)	3×10^4	924.03
		σ_1	3.97	(3.63, 4.34)	3×10^4	29038.93
	(c)	r	2.66	(2.53, 2.78)	3×10^4	2650.74
		σ_1	5.13	(4.80, 5.48)	3×10^4	29443.54
2	(a)	r_1	0.61	(0.57, 0.67)	3×10^4	2193.73
		r_2	6.41	(5.97, 6.97)	3×10^4	3303.21
		t^{critical}	2014.40	(2014.02, 2014.87)	3×10^4	3082.00
		σ_2	1.61	(1.50, 1.72)	3×10^4	28240.22
	(b)	r_1	1.31	(0.66, 1.49)	3×10^4	567.10
		r_2	8.12	(7.12, 9.25)	3×10^4	708.20
		t^{critical}	2017.01	(2015.61, 2017.70)	3×10^4	490.91
		σ_2	1.40	(1.28, 1.53)	3×10^4	25586.40

models and were explored after model training. Configuration (c) was explored in the early phase of model development, where we trained the monophasic Model 1 on the training data, including the ahead-of-front observations, to verify that the inferred expansion rate is robust to their removal. Summary statistics of the obtained posterior distributions are provided in Table A.4.

For configuration (a) (tighter initial profile) and for each model, we obtain 3×10^4 samples from the corresponding posteriors. For the monophasic Model 1, we obtain a posterior mode expansion rate of 2.78 and a posterior mode observational noise of 4.44. The decrease in the observational noise compared to that obtained in the main results (Table 2 - 4.81) is consistent with the hypothesis that these sites are identified as infested earlier due to overlapping with the initial condition, further supported by the expansion rate (2.78, up from 2.68) since the posterior mode for configuration (a) lies in the 95% credible interval in Table 2, implying little change in the inferred expansion. For the biphasic Model 2, we obtain a similar conclusion. There is little change in the inferred second phase expansion rate and transition time, and a small decrease in the observational noise compared to that obtained when a wider Gaussian is employed. Importantly, we still obtain distinct differences in the expansion rates for each phase, indicating the apparent non-constant rate of expansion is not an artefact of these observations. However, we do find that the first phase expansion rate increases when we tighten the initial profile, from 0.32 to 0.61. This could be due to the phase 1 expansion rate compensating for the larger distance the front must travel to reach a similar distance from the source at the end of phase 1 as it had reached under the wider initial profile.

For configuration (b) (removal of locations where the initial density is non-negligible) and for each model, we obtain 3×10^4 samples from the corresponding posteriors. For Model 1, the posterior mode expansion rate is 2.78 and the observational noise is 3.97. Compared to configuration (a), we obtain the same expansion rate estimate with lower observational noise, further supporting our claim that the observations within the region of non-negligible initial density are identified as infested earlier than they are observed to be. However, for Model 2, we observed a more substantial difference in the inferred posteriors; a phase 1 expansion rate of 1.31 (vs. 0.32 in Table 2), phase 2 expansion rate of 8.12 (vs 6.14), transition time of 2017.01 (vs. 2014.07), and observational noise parameter 1.40 (vs. 1.65). This suggests the removed observations have more impact on the inferred expansion under a biphasic model than they do under a monophasic model. However, we still find a distinct difference in the expansion rates of the two phases which assures us that the apparent non-constant rate of expansion is not an artefact of initial condition. One possible explanation for the difference in the posteriors under configuration (b) vs the main results is that the removed observations are all concentrated in phase 1. This leads to an increase in the phase 1 expansion rate, since there are fewer observations concentrated near the source in later years that would be otherwise require a slower moving front to obtain an accurate prediction of the arrival time. The model could respond by increasing the length of phase 1 as, under the faster expansion, sites with arrivals in 2014 to 2016 would now be reached by the front earlier. However, the expansion between 2014 and 2016 is slower under configuration (b) than under the main results, thus the model could compensate with a larger phase 2 expansion rate to ensure the front reaches observations in later years in adequate time.

In configuration (c) (including ahead-of-the-front observations), the Model 1 expansion rate of 2.66 is close enough to the previously obtained value of 2.68 to suggest little-to-no change in the inferred expansion of OPM, and the increase in observational noise (5.13 vs 4.81) is consistent with the assumption that the ahead-of-the-front observations are due to spread mechanisms not considered in these models, since the observational noise captures expected deviations away from the global model.

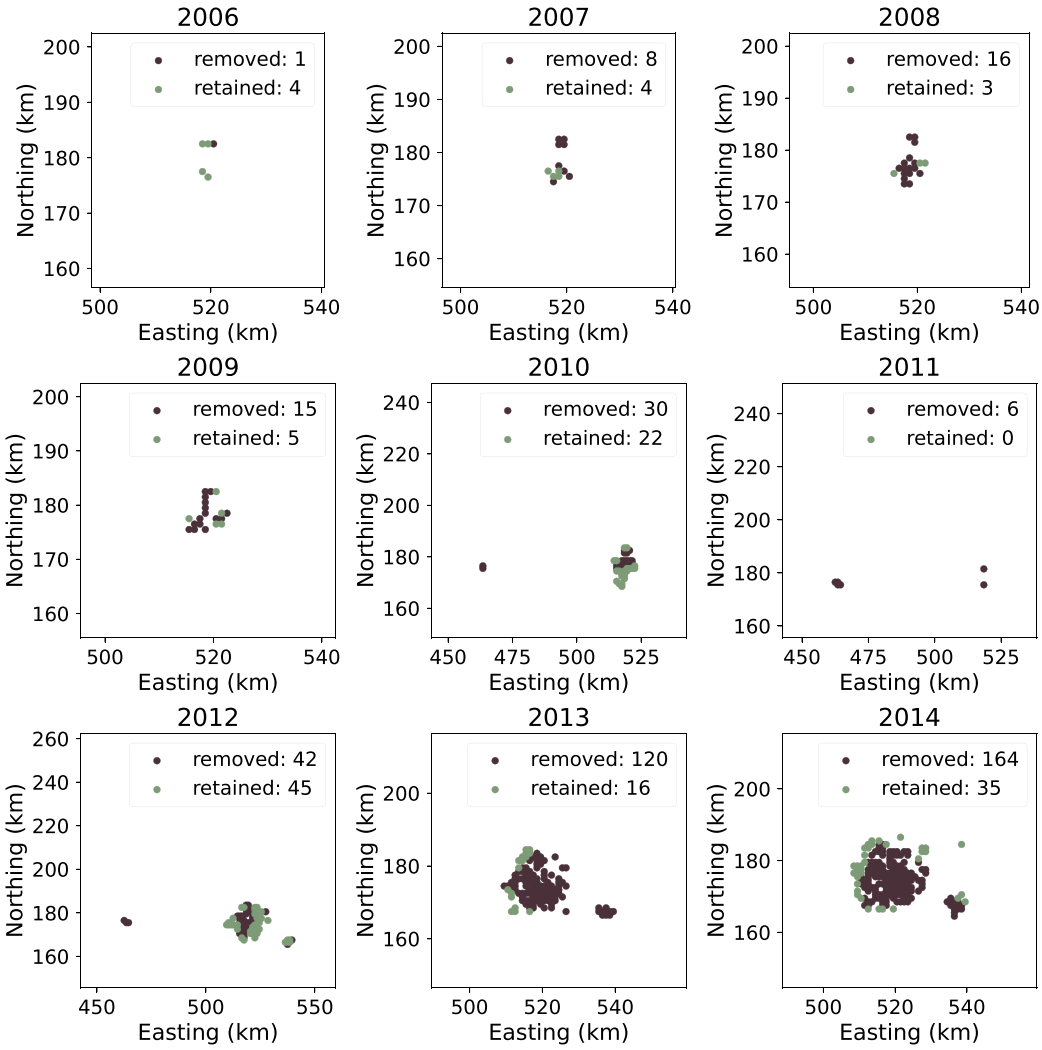


Fig. A.8. OPM observations, 2006 to 2014. The 1km^2 locations where OPM was observed in the respective year, shown as squares. Green squares are the sites that were retained as training data, while brown squares are those that were removed (see Table A.3 for a summary). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

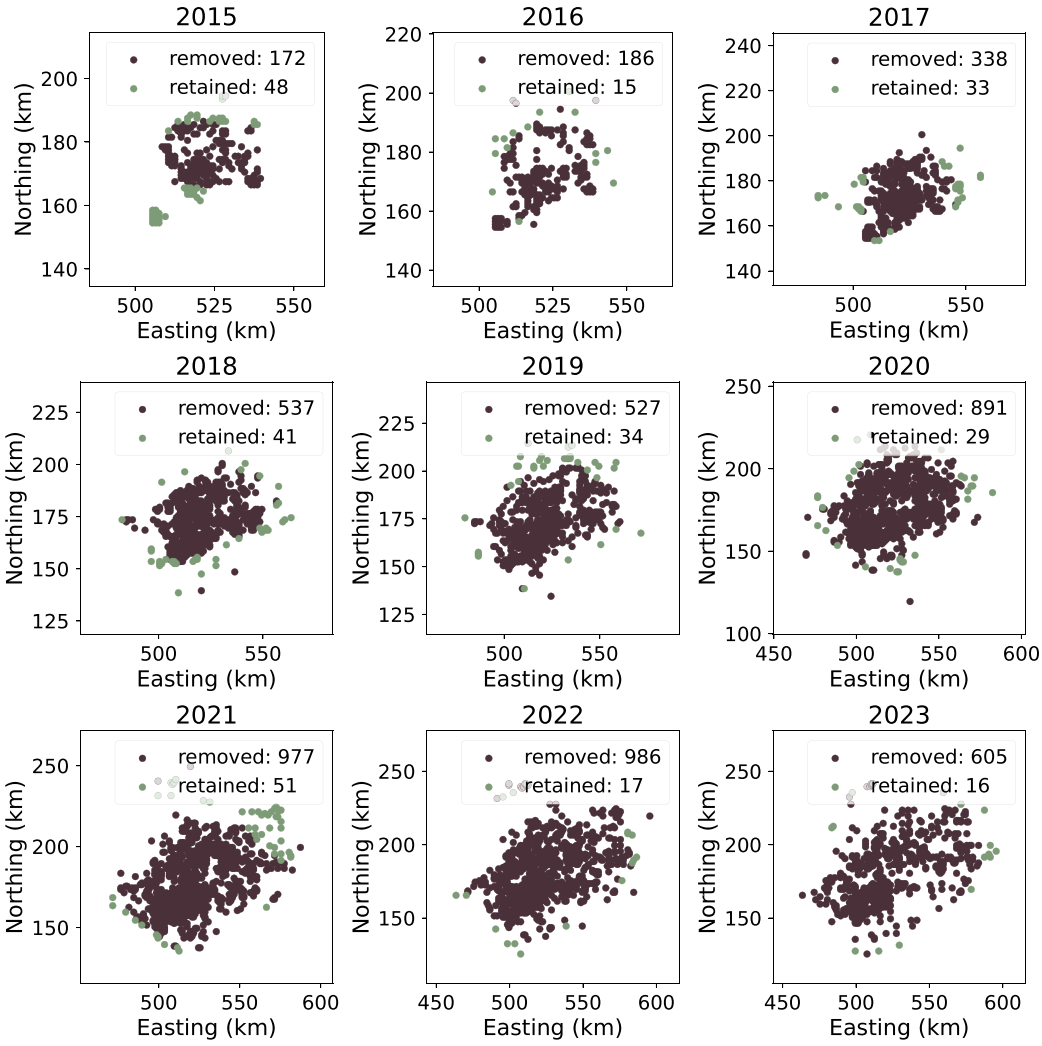


Fig. A.9. OPM observations, 2015 to 2023. The 1km^2 locations where OPM was observed in the respective year, shown as squares. Green squares are the sites that were retained as training data, while brown squares are those that were removed (see Table A.3 for a summary). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Appendix B. Sensitivity to the choice of prior distributions

In both reaction-diffusion models considered in this paper, we adopt the log-normal family as the prior distribution of (the components of) the model parameter θ and an inverse gamma distribution for the prior of the observational noise parameters σ. Here, we assess sensitivity of posterior inferences to the prior specification employed.

Since the prior means of each parameter are chosen to align with available estimates of the corresponding parameters in the literature, we fix these throughout the sensitivity analysis. For each model, we consider three additional prior specifications, where the variance is (a) one order of magnitude smaller (multiplying original variance by $A = 0.1$); (b) one order of magnitude larger ($A = 10$) and; (c) two orders of magnitude larger ($A = 100$) than those adopted in the main body (Section 2.4).

Recall that the growth parameter components r, r_1, and r_2 each have a prior distribution of $\text{LogN}(1.79, 0.79^2)$ and the transition time t^{critical} has a prior of $\text{LogN}(7.61, 0.006)$. For some $\lambda \sim \text{LogN}(\mu, \sigma^2)$, λ has mean $\bar{\lambda}_{\mu,\sigma} = \exp\left(\mu + \frac{\sigma^2}{2}\right)$ and variance $\delta(\lambda)_{\mu,\sigma} = \left[\exp(\sigma^2) - 1\right]\exp(2\mu + \sigma^2)$. We require $\bar{\mu}$, $\bar{v}$ such that $\bar{\lambda}_{\mu,\sigma} = \bar{\lambda}_{\bar{\mu},\bar{\sigma}}$ and $\delta(\lambda)_{\bar{\mu},\bar{\sigma}} = A\delta(\lambda)_{\mu,\sigma}$, where $A = 0.1, 10, 100$. Therefore,

$$\bar{\mu} = \mu + \frac{1}{2}(\sigma^2 - \bar{\sigma}^2) \tag{B.1}$$

$$\bar{\sigma}^2 = \log(1 + A(\exp(\sigma^2) - 1)). \tag{B.2}$$

Similarly, the squares of the observational noise parameters σ_1 and σ_2 have a prior distribution $\text{IG}(2.23, 3.72)$. Since some $\lambda \sim \text{IG}(\alpha, \beta)$ has mean $\bar{\lambda}_{\alpha,\beta} = \beta(\alpha - 1)^{-1}$ and variance $\delta(\lambda)_{\alpha,\beta} = \beta^2(\alpha - 1)^{-2}(\alpha - 2)^{-1}$, we require $\bar{\alpha}, \bar{\beta}$ such that

$$\bar{\alpha} = \frac{(\alpha - 2)}{A} + 2 \tag{B.3}$$

$$\bar{\beta} = \frac{\beta(\bar{\alpha} - 1)}{\alpha - 1}. \tag{B.4}$$

In Table B.5, we present the results of this prior sensitivity analysis (excluding the case $A = 1$, which is given in Table 2). In particular, for each model and each prior configuration, we provide the mean and 95% equi-tailed credible interval for the component chains. We find that our main results in Table 2 are robust to the choice of prior specification when the variance of the prior is increased (i.e. when we have less confidence *a priori*), but we obtain different estimates of the model parameters when the prior is tightened (higher confidence *a priori*). In particular, we find that increasing the region of high prior mass does

Table B.5
Table of results for sensitivity analyses of model outputs to the choice of prior distributions. For each additional prior configuration and each component parameter chain, we provide the mean, 95% equi-tailed credible intervals (CI), sample size (SS), and effective sample size (ESS).

Model	A	$\hat{}$	Mean	95% CI	SS	ESS
1	0.1	r	2.91	(2.78, 3.07)	3×10^4	2413.34
		σ_1	4.86	(4.54, 5.22)	3×10^4	17029.52
	10	r	2.68	(2.56, 2.81)	3×10^4	7237.99
		σ_1	4.81	(4.50, 5.16)	3×10^4	29851.40
	100	r	2.68	(2.56, 2.81)	3×10^4	987.43
		σ_1	4.82	(4.50, 5.16)	3×10^4	28612.85
2	0.1	r_1	0.35	(0.32, 0.38)	3×10^4	2202.21
		r_2	6.21	(5.83, 6.63)	3×10^4	2564.37
		t^{critical}	2014.16	(2013.84, 2014.49)	3×10^4	2387.37
		σ_2	1.66	(1.55, 1.78)	3×10^4	23059.35
	10	r_1	0.32	(0.28, 0.35)	3×10^4	3545.23
		r_2	6.13	(5.76, 6.53)	3×10^4	2615.63
		t^{critical}	2014.05	(2013.73, 2014.38)	3×10^4	2784.26
		σ_2	1.65	(1.54, 1.77)	3×10^4	28905.11
	100	r_1	0.32	(0.28, 0.35)	3×10^4	3496.60
		r_2	6.14	(5.77, 6.56)	3×10^4	2167.13
		t^{critical}	2014.05	(2013.73, 2014.39)	3×10^4	2091.93
		σ_2	1.65	(1.55, 1.77)	3×10^4	26851.21

Table C.6
Table of MCMC convergence results. For each of the two proposed models, we produced three MCMC chains with different initial values. For each component parameter chain, we provide the mean, 95% equi-tailed credible intervals (CI), sample size (SS), and effective sample size (ESS).

Model	$\hat{}$	Initial	Mean	95% CI	SS	ESS
1	r	4.0	2.68	(2.56, 2.81)	3×10^4	1803.75
	σ_1	1.0	4.81	(4.50, 5.15)	3×10^4	28508.55
	r	8.00	2.68	(2.56, 2.81)	3×10^4	4076.08
	σ_1	4.00	4.81	(4.50, 5.15)	3×10^4	29850.45
	r	0.40	2.68	(2.56, 2.81)	3×10^4	5052.61
	σ_1	0.10	4.81	(4.50, 5.15)	3×10^4	29249.23
2	r_1	4.00	0.32	(0.29, 0.35)	3×10^4	2622.82
	r_2	4.00	6.14	(5.77, 6.56)	3×10^4	2390.97
	t^{critical}	2010.00	2014.07	(2013.74, 2014.39)	3×10^4	2696.18
	σ_2	2.00	1.65	(1.54, 1.77)	3×10^4	28486.24
	r_1	3.00	0.32	(0.29, 0.35)	3×10^4	3179.00
	r_2	6.00	6.15	(5.77, 6.56)	3×10^4	2972.84
	t^{critical}	2008.00	2014.07	(2013.75, 2014.40)	3×10^4	3121.13
	σ_2	1.00	1.65	(1.54, 1.77)	3×10^4	28854.42
	r_1	6.50	0.32	(0.29, 0.35)	3×10^4	1920.77
	r_2	2.10	6.14	(5.78, 6.54)	3×10^4	2339.35
	t^{critical}	2019.00	2014.07	(2013.75, 2014.39)	3×10^4	4411.39
	σ_2	3.0	1.65	(1.55, 1.77)	3×10^4	29388.25

not influence the results, suggesting that the likelihood is highly informative.

Appendix C. MCMC convergence

The main results of this work, presented in Section 3, are posterior distributions for the parameters of the adopted models (and thus the expansion rates of OPM). These results are obtained from a single run of the MCMC scheme presented in Section 2.4, using one initial value: $(r_0, \sigma_{1,0}) = (4.0, 1.0)$ for Model 1 and $(r_{1,0}, r_{2,0}, t_0^{\text{critical}}, \sigma_{2,0}) = (4.0, 4.0, 2010.0, 2.0)$ for Model 2. Each run simulates a Markov chain which converges to the respective stationary distribution; here we verify convergence (by using multiple initial states for the chain).

For each model, we consider three initial states for the respective Markov chains with values chosen arbitrarily (including the above choices) and generate three Markov chains of length 3×10^4. The four additional initial states considered are $(8.0, 4.0)$ and $(0.4, 0.1)$ for Model 1, and $(3.0, 6.0, 2008.0, 1.0)$ and $(6.5, 2.1, 2019.0, 3.0)$ for Model 2. For each of the six chains, we calculate the mean and 95% equi-tailed credible intervals using the sample output. For each model, comparison of the mean and credible intervals (Table C.6), as well as visual comparison of the trace plots (Figs. C.10 and C.11), strongly imply that the Markov chains converge to the same stationary distribution. We also calculate the effective sample sizes (ESS) for each component chain (Table C.6), all of which exceed 1800 and suggest adequate mixing, which is also evidenced by the trace plots.

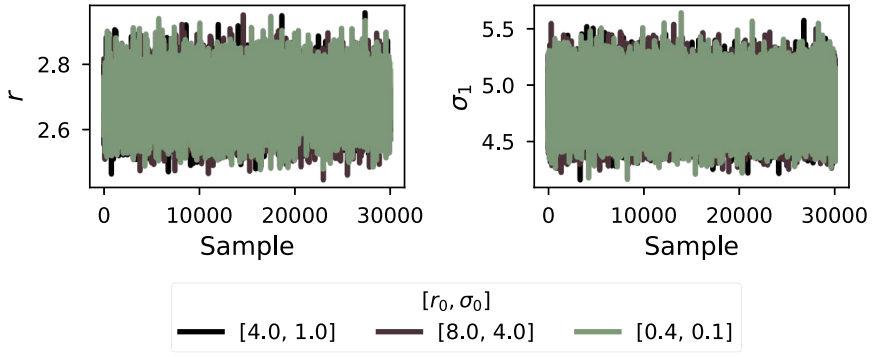


Fig. C.10. Model 1 trace plots. Trace plots based on the final MCMC run and two additional runs (see Table C.6), each with different initial values, for each parameter chain from the output of the MCMC scheme.

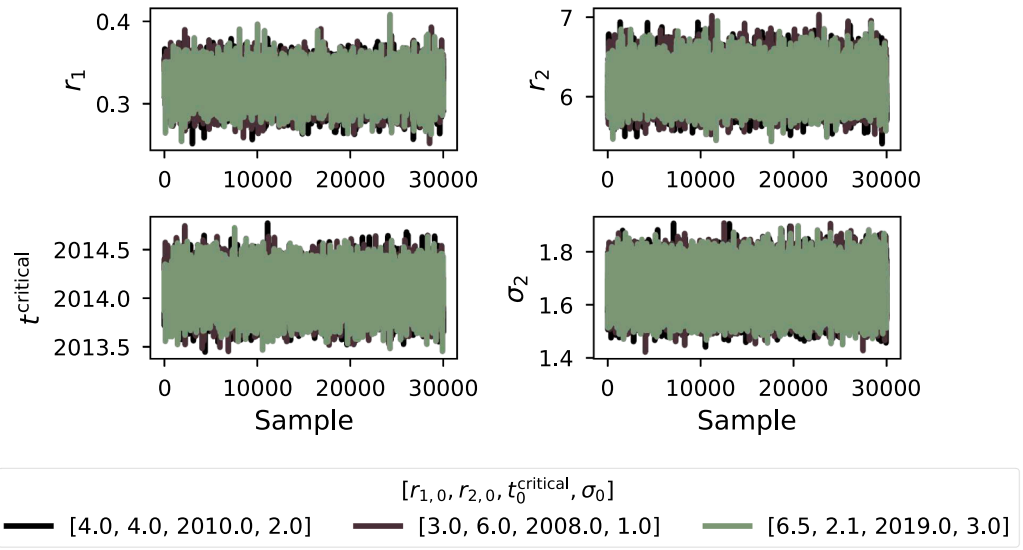


Fig. C.11. Model 2 trace plots. Trace plots based on the final MCMC run and two additional runs (see Table C.6), each with different initial values, for each parameter chain from the output of the MCMC scheme.

Appendix D. Radial models

Solutions to the monophasic and biphasic reaction-diffusion models used in this work evolve to a symmetric invasion front. Thus a radial model could capture the same pattern of spread as the reaction-diffusion models. Here, we train two linear radial models on the OPM observational data, one monophasic model and one biphasic model.

Recall (Section 2.4) that the likelihood function depends on the simulated arrival time $\tau(\vec{s}_k|\theta)$ with observational noise parameter σ. We define a monophasic radial model as follows. For each site $\vec{s}_k = (x_k, t_k)$, let E_k be the Euclidean distance between x_k and the centre of the initial states x^c. We set

$$\tau(\vec{s}_k|r) = \frac{E_k}{r} + t_0, \tag{D.1}$$

where $t_0 = 2006$ is the initial observational year, using the principle that (elapsed) time is distance over speed. We sample (r, σ) from their joint posterior distribution via the MCMC scheme outlined in Section 2.4, generating 3×10^4 samples from the target posterior. The (marginal) posterior distributions are shown in Fig. D.12. Additionally, we generate 10^4 posterior predictive samples for all sites, visualising the posterior predictive box plots for the nine selected sites (Table 1); these box plots are shown in Fig. D.13. Summary statistics, as well as the DIC for this model, are provided in Table D.7.

Likewise, we define a biphasic radial model, where the simulated arrival time at site $\vec{s}_k$ is

$$\tau\left(\vec{s}_k|r_1, r_2, t^{\text{critical}}\right) = \min\left[\frac{E_k}{r_1}, \bar{t} + \frac{E_k - r_1\bar{t}}{r_2}\right] + t_0. \tag{D.2}$$

where $\bar{t} = t^{\text{critical}} - t_0$. Since $r_1\bar{t}$ is the position of the front at time t^{critical} (cf. Eq. (D.1)), $(E_k - r_1\bar{t})/r_2$ is the elapsed time between t^{critical} and the time when the front reaches $\vec{s}_k$. We sample $(r_1, r_2, t^{\text{critical}}, \sigma)$ from their joint posterior distribution via the MCMC scheme outlined in Section 2.4, generating 3×10^4 samples from the target posterior. The (marginal) posterior distributions are shown in Fig. D.14 and box plots of the 10^4 posterior predictive samples for the nine selected sites (Table 1) are shown in Fig. D.15. Summary statistics and the DIC for this model are provided in Table D.7.

Table D.7
Table of results for the radial models. We provide, for each of the two proposed models, the calculated DIC as well as the mean, 95% equi-tailed credible interval (CI), sample size (SS), and effective sample size (ESS) for each of the model parameters.

Model	$\hat{}$	Mean	95% CI	SS	ESS	DIC
1	r	3.23	(3.12, 3.33)	3×10^4	4105.97	1491.21
	σ_1	3.65	(3.41, 3.91)	3×10^4	29194.97	
2	r_1	1.06	(0.98, 1.15)	3×10^4	2133.79	792.20
	r_2	6.51	(6.13, 6.97)	3×10^4	2620.63	
	t^{critical}	2013.87	(2013.47, 2014.32)	3×10^4	2718.35	
	σ_2	1.59	(1.48, 1.70)	3×10^4	28523.28	

The radial models provide the same overall conclusion on the spread of OPM – that the data prefer a model with non-constant expansion rate over one with a constant expansion rate, as indicated by the difference in DIC values for the monophasic radial model (1491.21) and the biphasic radial model (792.20), as well as the lower phase 1 expansion rate for the latter (1.06) vs the higher phase 2 expansion rate (6.51). We also find that each radial model marginally outperforms the corresponding reaction-diffusion model; for the monophasic models, the difference in the DICs is 229.53, whereas the difference for the biphasic models is 39.40, implying the best model of the four is the biphasic radial model. However, the credible intervals for the biphasic reaction-diffusion model parameters r_2, t_c, and σ_2 overlap with the corresponding credible intervals for the biphasic radial model. Thus the difference in the DICs for the two biphasic models is highly likely to be due to the more substantial difference in the phase 1 expansion rate predictions. The increase in the phase 1 expansion (1.06 from 0.32) matches the pattern observed in the sensitivity analysis of Appendix B, where we tightened the initial Gaussian profile and removed observations which overlap with the initial condition. Since the radial model does not include a Gaussian initial condition and distances are measured from a single point (the centre of the initial distribution), we would expect the radial wavefront to travel faster to capture the arrival at observations that overlap with the initial profile (as these observations would not be subject to artificially early arrival times due to the adopted initial condition).

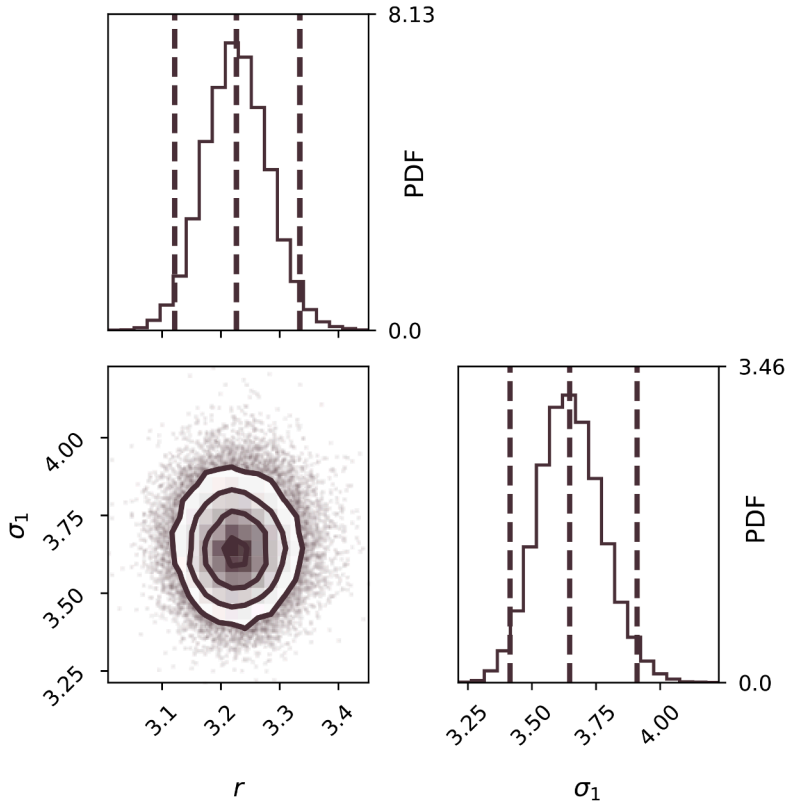


Fig. D.12. Monophasic radial model posterior distributions. Posterior distributions (histograms) and kernel density estimates of the pairwise marginal posteriors (contour plots) for r and σ_1, using 3×10^4 samples of the MCMC scheme. (Dashed) lines indicate the posterior mode and 95% credible interval.

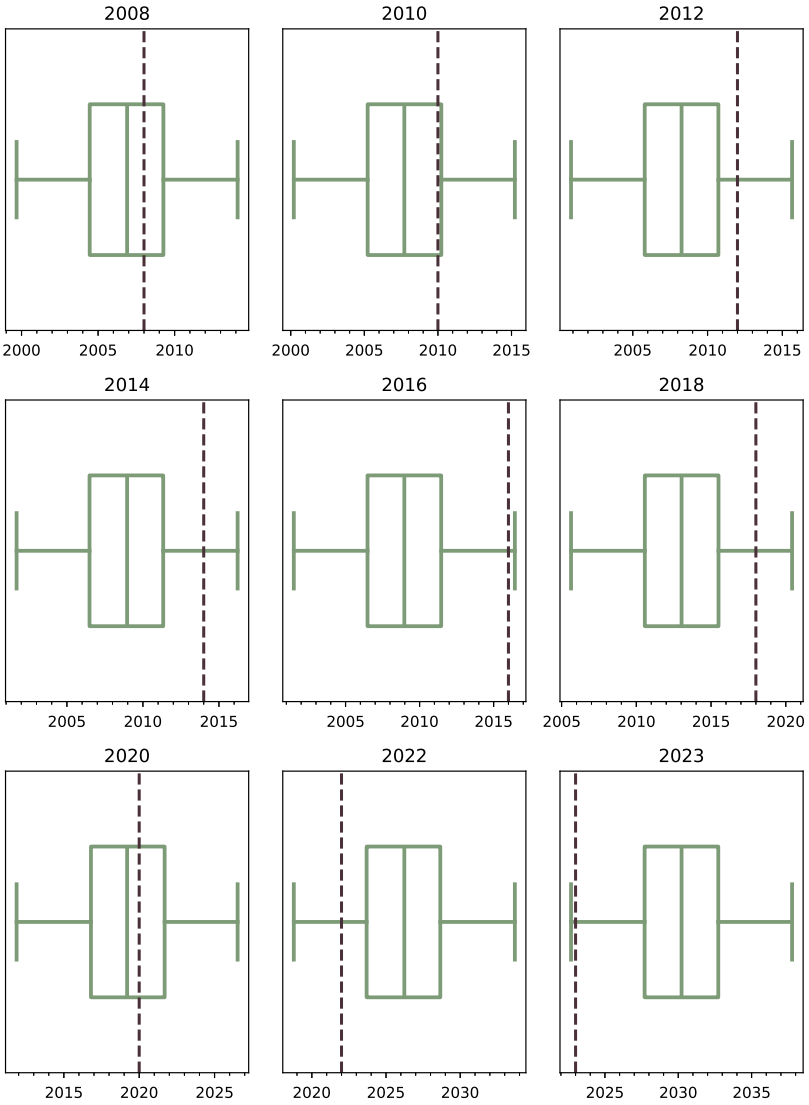


Fig. D.13. Monophasic radial model posterior predictives. Box plots summarising the predictive distribution at nine observed sites (test sites) (see Table 1). Whiskers indicate ±1.5IQR. Observations are indicated by a vertical dashed line, with the corresponding observed year above the axis.

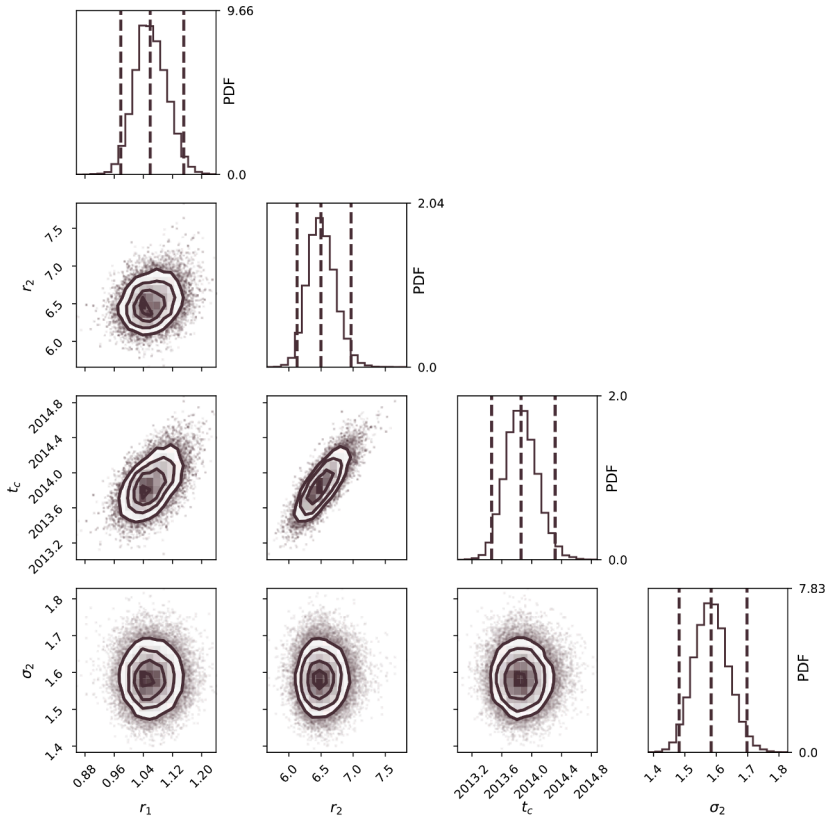


Fig. D.14. Biphasic radial model posterior distributions. Posterior distributions (histograms) and kernel density estimates of the pairwise marginal posteriors (contour plots) for r_1, r_2, t^{critical}, and σ_2, using 3×10^4 samples of the MCMC scheme. (Dashed) lines indicate the posterior mode and 95% credible interval.

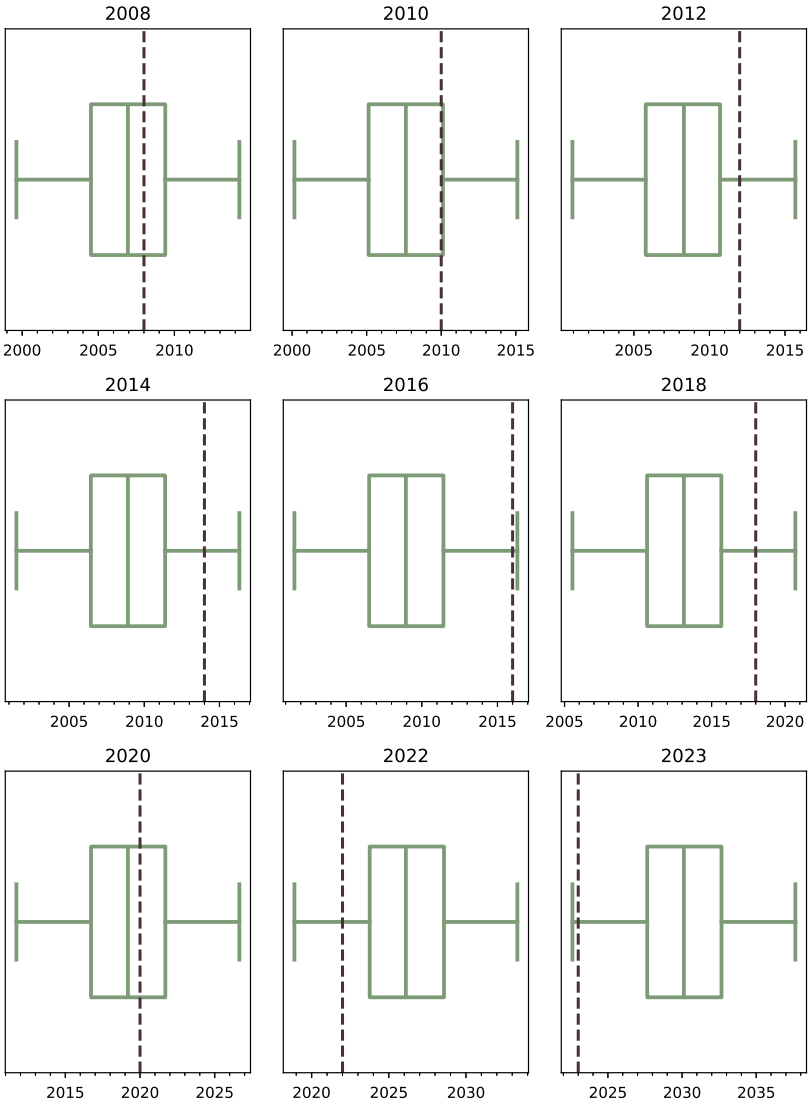


Fig. D.15. Biphasic radial model posterior predictives. Box plots summarising the predictive distribution at nine observed sites (test sites) (see Table 1). Whiskers indicate ±1.5IQR. Observations are indicated by a vertical dashed line, with the corresponding observed year above the axis.

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